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AIUB, Economics Notes

FINAL TERM

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MEASURING NATIONAL INCOME - GDP

Topic:1

Measuring National Income – GDP

Objectives

- **Output**: Achieve high-level and rapid growth of output.
- **Employment**: Maintain high levels of employment while minimizing involuntary unemployment.
- **Price-Level Stability**: Ensure stability in price levels to avoid inflation or deflation.

Instruments

1. **Monetary Policy**
    - Control money supply to influence interest rates.
2. **Fiscal Policy**
    - **Government Expenditure**: Boost economic activity through public spending.
    - **Taxation**: Adjust tax rates to influence disposable income and spending.

Macroeconomics

⇨ Macroeconomics looks at the economy as a whole.

Economy-wide phenomena, including inflation, unemployment, and economic growth.

Three Key Macroeconomic Statistics:

1. **GDP (Gross Domestic Product)**: Measures a country's total income and spending on goods and services.
2. **CPI (Consumer Price Index)**: Tracks the average price level of goods and services (shows inflation).
3. **Unemployment Rate**: Shows the percentage of people without jobs in the workforce.

Economic Models

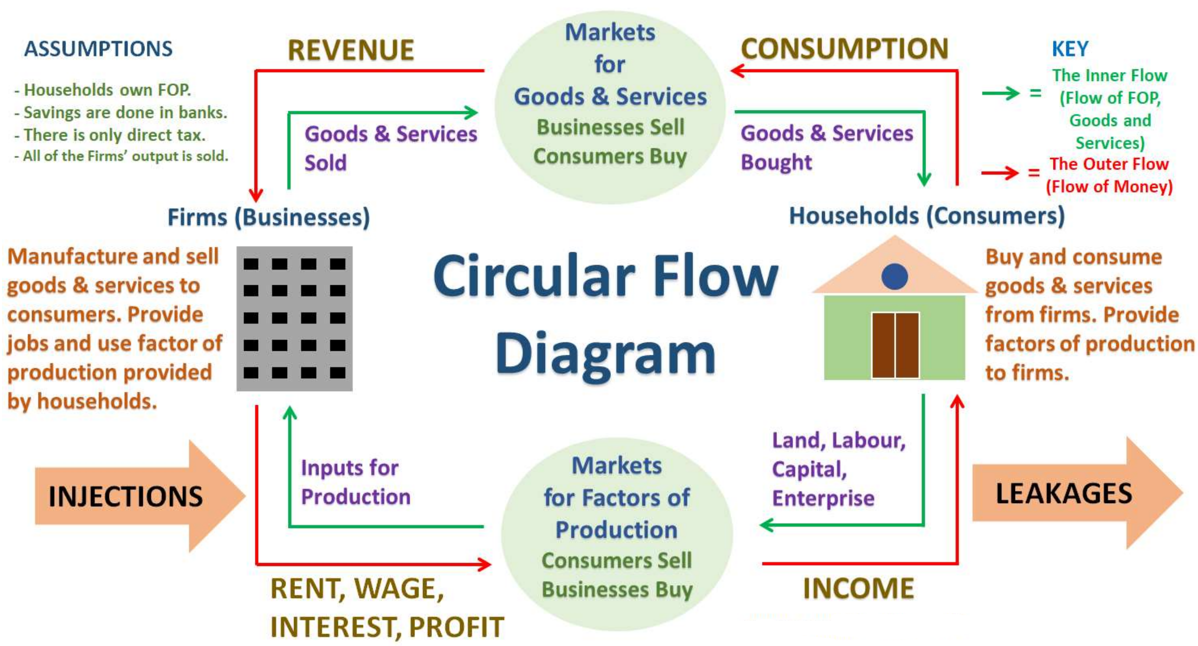
⇨ **Purpose**: Economists use models to simplify reality and better understand how the economy works.

Basic Macroeconomic Model

1. **The Circular Flow Diagram**:
    - Shows how money, goods, and services flow through the economy.
    - Includes:
        - **Households**: Provide labor and consume goods/services.
        - **Firms**: Produce goods/services and pay wages to households.
        - **Markets**: Facilitate the exchange of goods, services, and resources.
    - **Key Flows:**
        - **Money flows in one direction (wages, spending).**
        - **Goods, services, and labor flow in the opposite direction.**

Circular Flow Diagram

⇨ A visual presentation of the circular flow of **income in an economy** is called a circular flow diagram. This diagram illustrates the flow of **factors of production, outputs, and money in an economy.** The circular flow diagram is given below.



THE MEASUREMENT OF GROSS DOMESTIC PRODUCT (GDP)

➔ **Definition**: GDP is the market value of all final goods and services produced within a country in a specific period.

Key Points:

1. **"Market Value"**:
   - Output is measured at market prices.
2. **"Of All Final"**:
   - Includes only final goods, not intermediate goods (to avoid double counting).
   - Example: Paper used to make a Hallmark card is an intermediate good.
3. **"Goods and Services"**:
   - Includes **tangible** goods (e.g., food, clothing, cars).
   - Includes **intangible** services (e.g., haircuts, housecleaning, doctor visits).
4. **"Produced"**:
   - Counts only goods and services currently produced, not past transactions.
5. **"Within a Country"**:
   - Measures production within a country's borders, regardless of the producer's nationality.
6. **"In a Given Period of Time"**:
   - Reflects production within a specific timeframe, usually a year or a quarter.

Note: GDP includes all items produced in the economy and sold *legally* in markets.

Types of GDP

1. Nominal GDP
2. Real GDP

Nominal GDP (NGDP)

⇨ Nominal GDP values the production of goods and services at **current prices.**

Calculating Nominal Gross Domestic Product:

$$NGDP = \Sigma\, Quantity_{current\ year} * Price_{current\ year}$$

$$NGDP = \Sigma\, Q_{current\ year} * P_{current\ year}$$

Example:

Year 2015

Item	Quantity	Price
Goods	120	$50
Services	100	$65

Year 2016

Item	Quantity	Price
Goods	130	$55
Services	110	$70

a) Find the Nominal GDP for each year

⇨ Nominal GDP for Year 2015:

$$NGDP = \Sigma\, Q_{current\ year} * P_{current\ year}$$

$$\mathrm{NGDP}_{2015} = 120 * 50 + 100 * 65 = \$12,500$$

⇨ Nominal GDP for Year 2016:

$$NGDP = \Sigma\, Q_{current\ year} * P_{current\ year}$$

$$\mathrm{NGDP}_{2016} = 130 * 55 + 110 * 70 = \$14,850$$

Real GDP (RGDP)

⇨ Real GDP values the production of goods and services at **constant prices.**

Calculating Real Gross Domestic Product:

$$RGDP = \Sigma\, Quantity_{current\ year} * Price_{Base\ year}$$

$$RGDP = \Sigma\, Q_{current\ year} * P_{Base\ year}$$

Example:

Year 2015

Item	Quantity	Price
Goods	120	$50
Services	100	$65

Year 2016

Item	Quantity	Price
Goods	130	$55
Services	110	$70

a) Find the Real GDP for each year where 2015 is the base year.

⇨ **Real GDP for Year 2015:**

$$RGDP = \Sigma\, Q_{current\ year} * P_{Base\ year}$$

$$\mathbf{RGDP_{2015}} = \mathbf{120 * 50 + 100 * 65 = \$12,500}$$

⇨ **Real GDP for Year 2016:**

$$RGDP = \Sigma\, Q_{current\ year} * P_{Baase\ year}$$

$$\mathbf{RGDP_{2016}} = \mathbf{130 * 50 + 110 * 65 = \$13,650}$$

Notice that RGDP **has increased less** than NGDP from 2015 to 2016. This will always be the case if both prices and quantities increase from year to year.

Not Counted in GDP

➔**What Is Not Counted in GDP?**

1. **Non-Market Activities**:
   - Goods and services produced and consumed at home (e.g., home-cooked meals, unpaid childcare).
2. **Illegal Activities**:
   - Transactions involving illicit goods and services (e.g., illegal drugs).
3. **Second-Hand Sales**:
   - Sales of used goods (e.g., resale of cars, furniture).
4. **Financial Transactions**:
   - Purely financial activities like buying stocks or bonds.
5. **Transfer Payments**:
   - Government payments like pensions, welfare, or unemployment benefits.
6. **Intermediate Goods**:
   - Goods used as inputs for final goods (to avoid double counting).
7. **Environmental Costs**:
   - Negative impacts like pollution are not deducted.
8. **Volunteer Work**:
   - Services provided for free, even if they have economic value.

THE COMPONENTS OF GDP

GDP Formula: GDP (Y) **is calculated as the sum of:**

1. **Consumption (C)**: Spending by households on goods and services.
2. **Investment (I)**: Spending on capital goods, residential construction, and inventories.
3. **Government Purchases (G)**: Spending by the government on goods and services (excludes transfer payments).
4. **Net Exports (NX)**: Exports(X) minus imports(M)= (E-M)

Formula:

$$\boldsymbol{Y = C + I + G + NX}$$

$$\boldsymbol{Y = C + I + G + (X - M)}$$

1. Consumption (C)

- Spending by households on goods and services.
- **Excludes purchases** of new housing.

2. Investment (I)

- Spending on capital equipment, inventories, and structures.
- **Includes spending** on new housing.

3. Government Purchases (G)

- Spending on goods and services by local, state, and federal governments.
- **Excludes transfer payments (not exchanged for current goods or services).**

4. Net Exports (NX)

- **Exports(X) minus imports(M)= (E-M)**

GDP & Components of Expenditure

1. GDP

- Total value of goods and services produced in a country.

2. Consumption

- **Nondurable Goods**: Goods consumed quickly (e.g., **food**, **clothing**).
- **Durable Goods**: Long-lasting goods (e.g., **cars**, **appliances**).
- **Services**: Intangible goods (e.g., **healthcare**, **education**).

3. Investment

- **Nonresidential Fixed Investment**: Spending on **business equipment**, **machinery**, and **structures**.
- **Residential Fixed Investment**: Spending on **housing construction** and **improvements**.

4. Government Purchases

- **Defense**: Government spending on **military**.
- **Non-Defense**: Spending on other government services (e.g., **education, infrastructure**).

5. Net Exports

- **Exports**: Goods and services sold abroad.
- **Imports**: Goods and services purchased from abroad (subtracted from exports).

GDP deflator

⇨ The GDP deflator, also known as the **Implicit Price Deflator**, is an inflation indicator.

⇨ GDP deflator comes from nominal and real GDP

Formula:

$$GDP\ Deflator = \frac{Nominal\ GDP}{Real\ GDP} \times 100$$

$$GDPD = \frac{NGDP}{RGDP} \times 100$$

GDP Deflator and Price Levels

- **Purpose**:
  - The GDP deflator measures the **difference in price levels of goods and services** in a country over a specific timeframe.
- **Interpretation**:
  - It shows how much prices have changed, helping to assess inflation or deflation in the economy.

Example: 1

Year 2015

Item	Quantity	Price
Goods	120	$50
Services	100	$65

Year 2016

Item	Quantity	Price
Goods	130	$55
Services	110	$70

a) Find the Real GDP for each year where 2015 is the base year.
b) Find the Nominal GDP for each year
c) Find the GDP deflator for each year

(a)

⇨ **Real GDP for Year 2015:**

$$RGDP = \Sigma\, Q_{current\ year} * P_{Base\ year}$$

$$\mathbf{RGDP_{2015} = 120 * 50 + 100 * 65 = \$12,500}$$

⇨ **Real GDP for Year 2016:**

$$RGDP = \Sigma\, Q_{current\ year} * P_{Baase\ year}$$

$$\mathbf{RGDP_{2016} = 130 * 50 + 110 * 65 = \$13,650}$$

(b)

⇨ **Nominal GDP for Year 2015:**

$$NGDP = \Sigma\, Q_{current\ year} * P_{current\ year}$$

$$\mathbf{NGDP_{2015} = 120 * 50 + 100 * 65 = \$12,500}$$

⇨ **Nominal GDP for Year 2016:**

$$NGDP = \Sigma\, Q_{current\ year} * P_{current\ year}$$

$$\mathbf{NGDP_{2016} = 130 * 55 + 110 * 70 = \$14,850}$$

(c)

⇨ **GDP Deflator for Year 2015:**

$$GDPD = \frac{NGDP_{2015}}{RGDP_{2015}} \times 100$$

$$= \frac{12{,}500}{12{,}500} \times 100$$

= 100%

⇨ **GDP Deflator for Year 2016:**

$$GDPD = \frac{NGDP_{2016}}{RGDP_{2016}} \times 100$$

$$= \frac{14{,}850}{13{,}650} \times 100$$

= 108.79%

GDP AND ECONOMIC WELL-BEING

- **GDP as a Measure:**
    - GDP is the best single measure of a society's economic well-being.
- **GDP Per Person:**
    - GDP per person shows the average income and spending of individuals in the economy.
    - A higher GDP per person usually means a higher standard of living.
- **Limitations of GDP:**
    - GDP is not perfect because it doesn't include everything that contributes to well-being.
    - Some things that aren't counted in GDP include:
        - **Leisure**: The value of free time.
        - **Clean Environment**: The value of a healthy environment.
        - **Non-Market Activities**: Things done outside of the market, like time spent by parents with children or volunteer work.

✍ Problem 2:

➔ Below are the data for Pizza and Cake market.

Year	Price (Pizza)	Quantity (Pizza)	Price (Cake)	Quantity (Cake)
2010	9	200	20	150
2011	10	300	22	200
2012	12	300	24	200

a) Compute **nominal GDP**, **real GDP**, and the **GDP deflator** for each year, using **2010 as the base year.**

b) Compute the **percentage change in nominal GDP, real GDP, and the GDP deflator** in 2011 and 2012 from the preceding year.

c) What is the difference between nominal GDP and Real GDP?

d) Explain why real GDP is not a perfect measure of economic welfare.

e) Explain the expenditure approach of GDP and the circular flow of GDP.

(a)

⇨ **Real GDP for Year 2010:**

$$RGDP = \Sigma\, Q_{current\ year} * P_{Base\ year}$$

$$\mathbf{RGDP_{2010} = 200 * 9 + 150 * 20 = 4800}$$

⇨ **Real GDP for Year 2011:**

$$RGDP = \Sigma\, Q_{current\ year} * P_{Baase\ year}$$

$$\mathbf{RGDP_{2011} = 300 * 9 + 200 * 20 = 6700}$$

⇨ **Real GDP for Year 2012:**

$$RGDP = \Sigma\, Q_{current\ year} * P_{Baase\ year}$$

$$\mathbf{RGDP_{2012} = 300 * 9 + 200 * 20 = 6700}$$

⇨ **Nominal GDP for Year 2010:**

$$NGDP = \Sigma Q_{current\ year} * P_{current\ year}$$

$$\mathbf{NGDP_{2010} = 200 * 9 + 150 * 20 = 4800}$$

⇨ **Nominal GDP for Year 2011:**

$$NGDP = \Sigma Q_{current\ year} * P_{current\ year}$$

$$\mathbf{NGDP_{2011} = 300 * 10 + 200 * 22 = 7400}$$

⇨ **Nominal GDP for Year 2012:**

$$NGDP = \Sigma Q_{current\ year} * P_{current\ year}$$

$$\mathbf{NGDP_{2012} = 300 * 12 + 200 * 24 = 8400}$$

⇨ **GDP Deflator for Year 2010:**

$$GDPD = \frac{NGDP_{2010}}{RGDP_{2010}} \times 100$$

$$= \frac{4800}{4800} \times 100$$

= 100%

⇨ **GDP Deflator for Year 2011:**

$$GDPD = \frac{NGDP_{2011}}{RGDP_{2011}} \times 100$$

$$= \frac{7400}{6700} \times 100$$

= 110.44%

⇨ **GDP Deflator for Year 2012:**

$$GDPD = \frac{NGDP_{2012}}{RGDP_{2012}} \times 100$$

$$= \frac{8400}{6700} \times 100$$

= 125.37%

Summary:

Year	Nominal GDP	Real GDP	GDP Deflator
2010	4800	4800	100
2011	7400	6700	110.45
2012	8400	6700	125.37

(b)

⇨ **Percentage Change in nominal GDP for Year 2011:**

$$PCNGDP = \frac{NGDP_{2011} - NGDP_{2010}}{NGDP_{2010}} \times 100$$

$$= \frac{7400 - 4800}{4800} \times 100$$

= 54.167%

⇨ **Percentage Change in Real GDP for Year 2011:**

$$PCRGDP = \frac{RGDP_{2011} - RGDP_{2010}}{RGDP_{2010}} \times 100$$

$$= \frac{6700 - 4800}{4800} \times 100$$

= 39.583%

⇨ **Percentage Change in GDP Deflator for Year 2011:**

$$PCGDPD = \frac{GDPD_{2011} - GDPD_{2010}}{GDPD_{2010}} \times 100$$

$$= \frac{110.45 - 100}{100} \times 100$$

$$= 10.45\%$$

⇨ **Percentage Change in nominal GDP for Year 2012:**

$$PCNGDP = \frac{NGDP_{2012} - NGDP_{2011}}{NGDP_{2011}} \times 100$$

$$= \frac{8400 - 7400}{7400} \times 100$$

$$= 13.51\%$$

⇨ **Percentage Change in Real GDP for Year 2012:**

$$PCRGDP = \frac{RGDP_{2012} - RGDP_{2011}}{RGDP_{2011}} \times 100$$

$$= \frac{6700 - 6700}{6700} \times 100$$

$$= 0\%$$

⇨ **Percentage Change in GDP Deflator for Year 2012:**

$$PCGDPD = \frac{GDPD_{2012} - GDPD_{2011}}{GDPD_{2011}} \times 100$$

$$= \frac{125.37 - 110.45}{110.45} \times 100$$

$$= 13.50\%$$

Summary:

Year	Nominal GDP % Change	Real GDP % Change	GDP Deflator % Change
2011	54.17%	39.58%	10.45%
2012	13.51%	0%	13.50%

(c)

➔ Difference Between Nominal GDP and Real GDP

- **Nominal GDP** is calculated using current prices in each year. It does not account for inflation, meaning it can be influenced by changes in prices as well as changes in the quantity of goods and services produced.
- **Real GDP** is adjusted for inflation by using the prices from a base year. It reflects the actual increase in production of goods and services, without the distortion caused by price changes. It is a better measure for comparing economic output over time because it isolates changes in the volume of production.

(d)

➔ Why Real GDP is Not a Perfect Measure of Economic Welfare

Real GDP is not a perfect measure of economic welfare for several reasons:

- **Non-Market Activities**: It does not account for non-market activities like household labor or volunteer work, which can contribute significantly to welfare.
- **Income Distribution**: Real GDP does not reflect income inequality. A rise in GDP could be accompanied by a greater concentration of wealth among the rich, which may not improve overall welfare.
- **Environmental Degradation**: It does not account for negative externalities, like pollution or depletion of natural resources, which can reduce welfare.
- **Quality of Life**: It does not consider factors like leisure, health, and education, which contribute to well-being.
- **Changes in Social or Political Conditions**: It overlooks the impact of social or political factors that affect life quality, like crime or governance.

(e)

➔ The Expenditure Approach of GDP and the Circular Flow of GDP

- **Expenditure Approach**: This is one of the most common methods for calculating GDP. It sums up all expenditures made in an economy, usually divided into four main categories:
    1. **Consumption (C)**: Expenditure by households on goods and services.
    2. **Investment (I)**: Expenditure on capital goods like machinery and buildings.
    3. **Government Spending (G)**: Expenditure by the government on goods and services.
    4. **Net Exports (NX)**: Exports minus imports.

  The GDP formula using the expenditure approach is:

$$\boldsymbol{Y = C + I + G + (X - M)}$$

- **Circular Flow of GDP**: The circular flow model demonstrates how money moves through an economy. Households supply labor and other resources to firms and receive income (wages, rents, profits). They then use this income to purchase goods and services from firms, creating a continuous flow of money. The government and foreign sectors also interact with this flow. The model shows the interdependence of different sectors and how they contribute to the overall economy.

CONSUMER PRICE INDEX (CPI)

Topic: 2

THE CONSUMER PRICE INDEX

➔ When the CPI rises, **families spend more** to maintain their standard of living.

How the Consumer Price Index (CPI) Is Calculated

- **Fix the Basket:** Identify key goods and services in a typical consumer's spending.
- **Find the Prices:** Track basket item prices over time.
- **Compute the Basket's Cost:** Calculate the basket's total cost over time.

$$\text{Cost of Basket in Year1} = \sum \textit{product price in Year}1 \times \textit{product Quantity in year } 1$$

- **Choose a Base Year and Compute the Index:** Select a base year, then calculate the index by dividing the basket's price each year by the base year's price and multiplying by 100.

$$\text{CPI} = \frac{\text{Cost of the Market Basket in Given Year}}{\text{Cost of the Market Basket in Base Year}} \times 100\%$$

How the Inflation Rate Is Calculated

- **Compute the Inflation Rate:** Calculate the percentage change in the price index from the previous period.

$$\text{Inflation Rate} = \frac{\text{New CPI} - \text{Prior CPI}}{\text{Prior CPI}} \times 100$$

Problem1:

⇨ The basket of the economy consists of **10 unit of DVD and 13 unit of Laptop.**

Year	Price of DVD	Price of Laptop
2007 (Base Year)	50	20
2008	30	32
2009	90	44

a) Compute the CPI of each year.
b) Compute the inflation rate for 2008 and 2009.

(a)

⇨ **We Know:**

$$CPI = \frac{Cost\ Of\ Basket\ in\ Given\ Year}{Cost\ of\ the\ Basket\ in\ Base\ Year} \times 100$$

Step 1: Calculate the cost of the Basket for each year:

∴ **Cost of Basket in 2007 =** $10 \times 50 + 13 \times 20 = 760$

∴ **Cost of Basket in 2008 =** $10 \times 30 + 13 \times 32 = 716$

∴ **Cost of Basket in 2009 =** $10 \times 90 + 13 \times 44 = 1472$

Step 2: Calculate the CPI for each year:

$$\therefore CPI_{2007} = \frac{\text{Cost of Basket in 2007}}{\text{Cost of Basket in 2007 (Base Year)}} \times 100$$

$$\therefore CPI_{2007} = \frac{760}{760} \times 100\% = 100$$

$$\therefore CPI_{2008} = \frac{\text{Cost of Basket in 2008}}{\text{Cost of Basket in 2007 (Base Year)}} \times 100$$

$$\therefore CPI_{2008} = \frac{716}{760} \times 100\% = 94.21$$

$$\therefore CPI_{2009} = \frac{\text{Cost of Basket in 2009}}{\text{Cost of Basket in 2007 (Base Year)}} \times 100$$

$$\therefore CPI_{2009} = \frac{1472}{760} \times 100\% = 193.68$$

(b)

⇨ **We Know:**

$$Inflation\ Rate = \frac{CPI\ in\ Current\ Year - CPI\ in\ Previous\ Year}{CPI\ in\ Previous\ Year} \times 100\%$$

➔ **Inflation Rate for 2008:**

$$\therefore Inflation\ Rate = \frac{CPI\ in\ 2008 - CPI\ in\ 2007}{CPI\ in\ 2007} \times 100\%$$

$$\therefore Inflation\ Rate = \frac{94.21-100}{100} \times 100\% = -5.79\%$$ (deflation)

➔ **Inflation Rate for 2009:**

$$\therefore Inflation\ Rate = \frac{CPI\ in\ 2009 - CPI\ in\ 2008}{CPI\ in\ 2008} \times 100\%$$

$$\therefore Inflation\ Rate = \frac{193.68 - 94.21}{94.21} \times 100\% = 105.58\%$$

CPI Vs GDP deflator

➔ Here's the key differences between the **GDP deflator** and **CPI**:

1. **Coverage of Goods:**
   - **GDP deflator** measures the prices of all goods and services **produced domestically.**
   - **CPI** measures the prices of goods and services bought by consumers, **including imports.**
2. **Treatment of Imports:**
   - **GDP deflator** excludes imported goods.
   - **CPI** includes imported goods.
3. **Basket of Goods:**
   - **GDP deflator** allows the basket of goods to change over time based on production changes.
   - **CPI** uses a fixed basket of goods.

Problems in Measuring the Cost of Living- CPI Biases

➔ The **CPI** accurately measures the price changes of a typical basket of goods but is not a perfect measure of the cost of living.

➔ **Key problems in measuring the cost of living are:**

- Substitution bias
- Introduction of new goods

Substitution bias

➔ **Substitution bias** occurs because the CPI uses a fixed basket of goods, not reflecting consumer substitution toward cheaper alternatives when relative prices change. This causes the index to overstate the cost-of-living increase.

Introduction of new goods

➔**The introduction of new goods** increases variety, making each dollar more valuable and reducing the amount of money needed to maintain a given standard of living. However, the basket used in price indices does not reflect this change in purchasing power.

➔ Substitution bias and the introduction of new goods cause the CPI to overstate the true cost of living, which is important because many government programs use the CPI to adjust for price changes.

Nominal & Real Interest rate

➔The nominal interest rate is the reported rate, **not adjusted for inflation**, while the real interest rate is the nominal rate **adjusted for inflation.**

∴ Real Interest rate = Nominal interest rate – Inflation rate

INFLATION

➔**Inflation is the rise in the overall price level of the economy.**

- The inflation rate is the percentage change in price levels from the previous period.
- Not all prices rise at the same rate during inflation.
- Not everyone suffers equally from inflation.
- Inflation may harm some people but benefit others.
- Hyperinflation is an extraordinarily high inflation rate, exceeding 50% per month.
- Germany experienced hyperinflation in the 1920s.

Types of Inflation

- Demand-Pull Inflation
- Cost-Push Inflation

Demand-Pull Inflation

- Demand-pull inflation arises from **excessive demand** in the economy.
- It occurs when "**too much money chases too few goods**," allowing producers to increase prices.

Cost-Push Inflation

- Cost-push inflation **originates from the supply side.**
- Higher production costs drive up product prices.

Effects of Inflation:

1. **Zero Inflation**:
   - **Consumers**: Not affected.
   - **Producers**: No reason to produce more.
   - **Economy**: Stays the same, no growth.
2. **Mild Inflation**:
   - **Consumers**: A little affected, still buying as usual.
   - **Producers**: Motivated to produce more / Make more things because of demand.
   - **Economy**: Grows as production increases.
3. **High Inflation**:
   - **Consumers**: Strongly affected, demand drops.
   - **Producers**: Production decreases due to low demand.
   - **Economy**: Shrinks, less is produced.

Problem 2:

The basket of the economy consists of **10 unit of DVD and 13 unit of Laptop.**

Year	Price of DVD	Price of Laptop
2007 (Base Year)	50	20
2008	30	32
2009	90	44

a) Compute the CPI of each year.
b) Compute the inflation rate for 2008 and 2009.
c) What are the key differences between CPI and GDP deflator?
d) Explain why CPI overstate inflation.
e) Explain how CPI is measured.

(a)

We Know:

$$CPI = \frac{Cost\ Of\ Basket\ in\ Given\ Year}{Cost\ of\ the\ Basket\ in\ Base\ Year} \times 100$$

Step 1: Calculate the cost of the Basket for each year:

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$$\therefore CPI_{2009} = \frac{1472}{760} \times 100\% = 193.68$$

(b)

⇨ **We Know:**

$$Inflation\ Rate = \frac{CPI\ in\ Current\ Year - CPI\ in\ Previous\ Year}{CPI\ in\ Previous\ Year} \times 100\%$$

➔ **Inflation Rate for 2008:**

$$\therefore Inflation\ Rate = \frac{CPI\ in\ 2008 - CPI\ in\ 2007}{CPI\ in\ 2007} \times 100\%$$

$$\therefore Inflation\ Rate = \frac{94.21 - 100}{100} \times 100\% = -5.79\% \text{(deflation)}$$

➔ **Inflation Rate for 2009:**

$$\therefore Inflation\ Rate = \frac{CPI\ in\ 2009 - CPI\ in\ 2008}{CPI\ in\ 2008} \times 100\%$$

$$\therefore Inflation\ Rate = \frac{193.68 - 94.21}{94.21} \times 100\% = 105.58\%$$

(c)

➔ Key Differences Between CPI and GDP Deflator

CPI	GDP Deflator
Measures the cost of a fixed basket of goods.	Measures prices of all goods produced domestically.
Includes imported goods.	Excludes imported goods.
Fixed weights are used.	Flexible weights (adjust to production).

(d)

➔ CPI overstates inflation because:

1. **Substitution Bias**: It assumes consumers buy the same goods even if prices rise, ignoring substitutions.
2. **Quality Improvement**: It doesn't fully adjust for improvements in product quality.
3. **Introduction of New Goods**: CPI does not capture the benefits of new goods and services right away.

(e)

1. **Choose a basket of goods**: Determine a fixed set of items based on consumer spending patterns.
2. **Collect prices**: Record the prices of items in the basket over time.
3. **Calculate the cost of the basket**: Multiply quantities by prices to find the total cost.
4. **Compare with the base year**: Use the base year cost as a reference to calculate CPI using the formula:

$$\text{CPI} = \frac{\text{Cost of the Market Basket in Given Year}}{\text{Cost of the Market Basket in Base Year}} \times 100\%$$

Business Cycle and Unemployment

Topic: 3

Macroeconomic Instability

⇨ **Macroeconomic Instability** is when key economic factors like growth, inflation, or unemployment become unstable, harming confidence and growth.

- Economic Growth
- Business Cycle
- Unemployment
- Inflation

Economic Growth

⇨ Economic growth is the rise in real GDP or real GDP per capita over time.
⇨ If real GDP grows from $200 billion to $210 billion, the growth rate is 5%.

Rule of 70:

$$\textit{Years to double real GDP} = \frac{70}{\textit{annual percentage rate of growth}}$$

Example:

- **RGDP 2021** = 80 million dollars
- **RGDP 2022** = 100 million dollars

Question: How long will it take to double the RGDP?

Solution:

1. **Calculate Growth Rate**:

$$\textit{Growth Rate} = \frac{100 - 80}{80} \times 100\% = 25\%$$

2. **Apply Rule of 70**:

$$\textit{Approximate Years} = \frac{70}{\textit{Growth Rate}} = \frac{70}{25} = 2.8$$

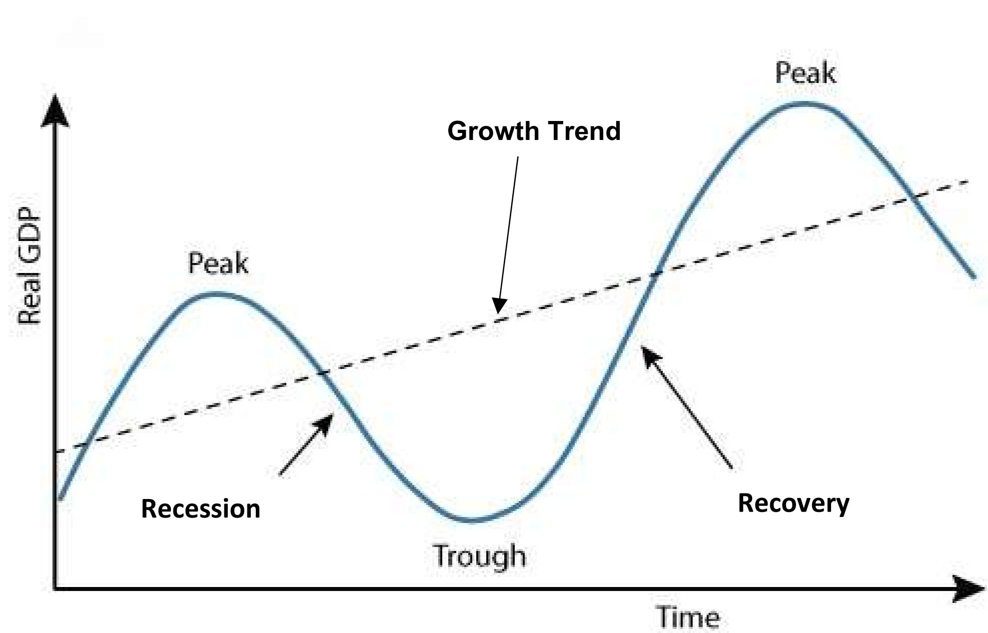
So, it will take approximately **2.8 years** to double the RGDP.

The Business Cycle

➔ The **business cycle** is the ups and downs in economic activity, including periods of growth and recession.

- **Business Cycle**: Alternating periods of economic growth and decline.
- **GDP Decline**: Leads to higher unemployment.
- **Rapid Growth**: Often accompanied by high inflation.

Stages of the Business Cycle



Peak

⇨ **Peak**: The economy hits a maximum, with full employment, near full productivity, and rising prices.

- **Economy reaches a temporary maximum.**
- **Full employment is achieved.**
- **Near full productivity.**
- **Prices tend to rise.**

Recession

⇨ A **recession** is a period of declining output, income, employment, and trade, marked by widespread business contraction and falling prices.

- **Drop in production, income, jobs, and trade.**
- **Many businesses shrink.**
- **Prices tend to fall.**

Trough

- **The lowest point for jobs and production.**
- **Can be short or long.**

Recovery

- **Recovery: output and jobs rise.**
- **Prices may rise before full capacity.**

Unemployment

➔The labor force is made up of people **who can work and want to work**, including those **with jobs and those looking for work.**

BLS divides population into 3 groups

- **Employed**: paid workers, self-employed, and unpaid family business workers.
- **Unemployed**: people available to work and actively looking in the past 4 weeks.

- **Not in the labor force:** those not working or seeking work, like students, homemakers, or retirees.

Note: The labor force includes both employed and unemployed workers.

Labor Force Statistics

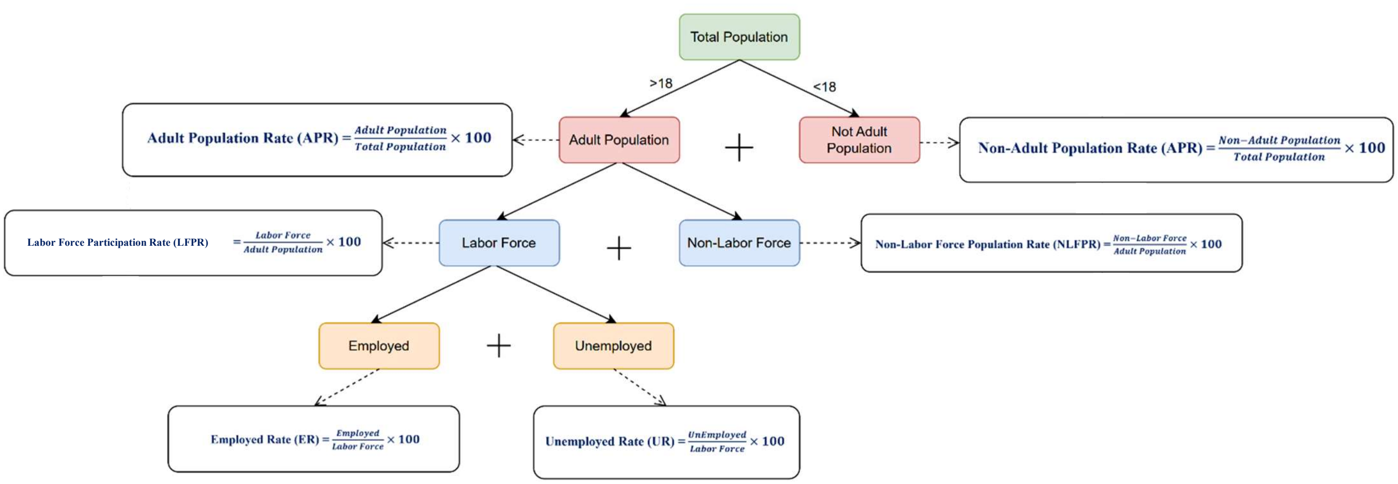
- Unemployment rate (u-rate): **% of the labor force that is unemployed.**

$$Unemployment\ rate = \frac{Number\ of\ unemployed}{Labor\ force} \times 100$$

- Labor force participation rate: **% of the adult population in the labor force.**

$$Labor\ force\ participation\ rate = \frac{Labor\ force}{Adult\ population} \times 100$$

Formulas:



Problem 1:

Adult population of the U.S. by group, June 2008	
# of employed	145.9 million
# of unemployed	8.5 million
not in labor force	79.2 million

➔ Compute the

a) Labor force,
b) U-rate,
c) Adult population, and
d) Labor force participation rate using this data

(a)

➔ We Know,

$$Labor\ Force = Employed + Unemployed$$

$$\therefore Labor\ Force = \mathbf{145.9 + 8.5 = 154.4}\ \boldsymbol{million}$$

(b)

➔ We Know,

$$UR = \frac{Unemployed}{Labor\ Force} \times 100\%$$

$$\therefore UR = \frac{\mathbf{8.5}}{\mathbf{154.4}} \times \mathbf{100\% = 5.50\%}$$

(c)

➔ We Know,

$$Adult\ Population = Labor\ Force + Non - Labor\ Force$$

$$\therefore Adult\ Population = 154.4 + 79.2 = 233.6\ million$$

(d)

➔ We Know,

$$LFPR = \frac{Labor\ Force}{Adult\ Population} \times 100\%$$

$$\therefore LFPR = \frac{154.4}{233.6} \times 100\% = 66.09\%$$

Types of Unemployment

- **Frictional**: Temporary, due to job searching or transitions.
- **Structural**: Mismatch of skills or locations with jobs available.
- **Cyclical**: Caused by economic downturns.

Frictional Unemployment

- Includes **search unemployment** and **wait unemployment**.
- Occurs due to imperfections in matching workers with jobs.

➔ Why Frictional Unemployment is inevitable?

Sectoral Shifts:

- **Changes in demand** across industries or regions.
- Some workers lose jobs and need to find new ones that fit their skills.
- Lead to inevitable frictional unemployment due to constant economic change.

Structural Unemployment

- Happens when **workers' skills or locations don't match new job opportunities**.
- Workers may need **retraining**, more **education**, or moving to find jobs.

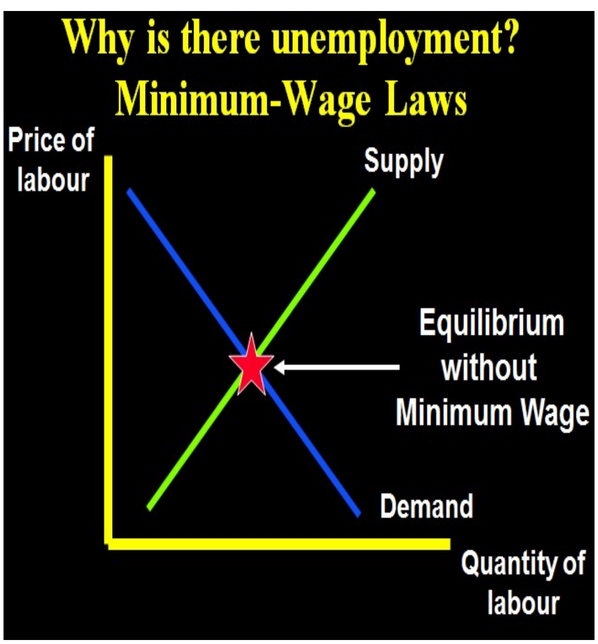
Why there is Structural unemployment?

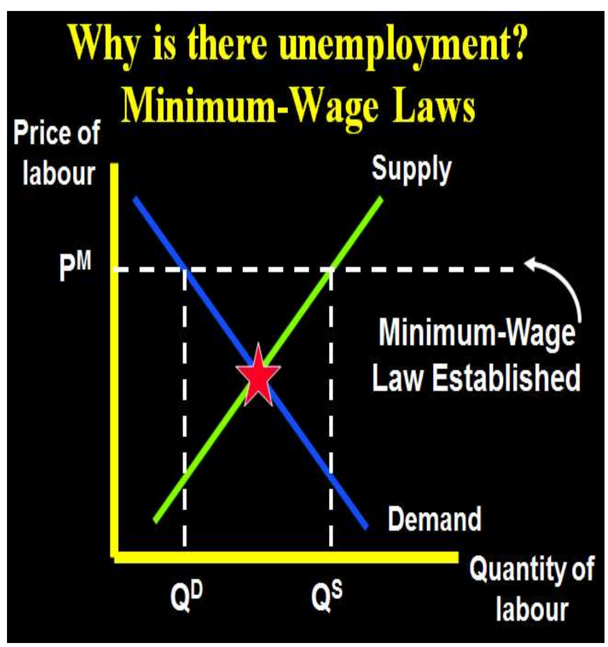
⇨ Happens when wages are higher than the market level. (equilibrium)

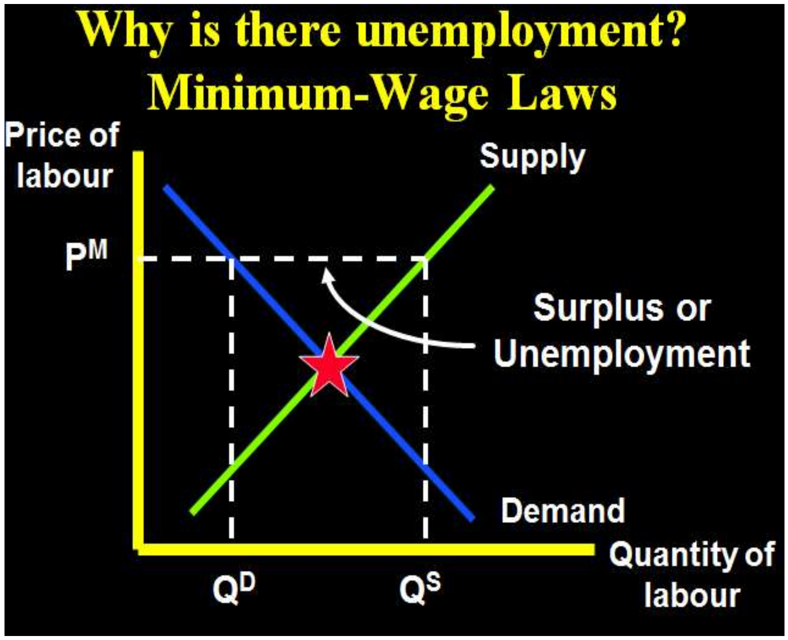
- Key causes:
  1. **Minimum wage laws**
  2. **Unions**
  3. **Efficiency wages**

Minimum wage laws

➔ When a minimum wage **is set higher than the market balance**, it leads to too many workers looking for jobs but not enough jobs available, creating unemployment, **called labor surplus.**







Unions

- A union is a group of workers who talk to employers about pay and working conditions.
- Unions work like a team to get better wages.
- Union workers make 20% more money and get more benefits than non-union workers doing the same job.

Employment situation under union:

- When **unions raise** wages above the balance point, fewer jobs are available, causing **unemployment.**
- **"Insiders"** are workers who **keep their jobs and benefit**.
- **"Outsiders"** are workers who **lose their jobs and are worse off.**
- **Some outsiders move to non-union jobs**, increasing labor supply and lowering wages in those markets.

Are unions good or bad for the economy?

Critics:

- **Unions are like cartels.**
- They raise wages too high, causing unemployment and lower wages in non-union jobs.

Advocates:

- Unions balance the power of big companies.
- They make companies pay more attention to workers' needs.

Efficiency Wages

Efficiency wages theory:

- **Firms work better** if they **pay higher wages than the equilibrium level.**
- They pay more to improve worker productivity.
- However, with unions and minimum wage laws, firms are forced to pay higher wages.

Four reasons firms pay efficiency wages:

1. **Worker health:** Higher wages help workers stay healthy and productive.
2. **Worker turnover:** Higher wages encourage workers to stay, reducing hiring and training costs.
3. **Worker quality:** Paying more attracts better job applicants and improves workforce quality.
4. **Worker effort:** When wages are above equilibrium, workers are motivated to work harder since jobs are scarce.

Cyclical Unemployment

➔ Cyclical unemployment happens when **total spending drops**, **usually during a recession.**

Problem 2:

a) Consider in an economy the **total population is 170 million** and **70% of the total population is not under adult population**. **Not in labor population is 17 million** and **total unemployed people are 18 million**. Calculate **labor force participation rate and total employment rate.**

b) Define marginally attached workers and part time works who want full time jobs

c) Explain different types of unemployment.

(a)

➔ Is Given,

Total Population	170 million
Non-Adult population rate (NAPR)	70%
Non-Labor Force	17 million
Unemployed	18 million

➔ We Know,

$$NAPR = \frac{NonAdult\ Population}{Total\ Population}$$

$$\therefore NonAdult\ Population = NAPR \times\ Total\ Population$$

$$\therefore NonAdult\ Population = 70\% \times\ 170 = 119\ million$$

$$\therefore Adult\ Population = Total\ Population -\ NonAdult\ Population$$

$$\therefore Adult\ Population = 170 - 119 = 51\ million$$

$$\therefore Labor\ Force = Adult\ Population -\ NonLabor\ Force$$

$$\therefore Labor\ Population = 51 - 17 = 34\ million$$

$$\therefore \textit{Employment} = \textit{Labor Force} - \textit{Unemployed}$$

$$\therefore \textit{Employment} = 34 - 18 = 16 \textit{ million}$$

So, Labor Force Participation Rate (LFPR):

$$LFPR = \frac{\textit{Labor Force}}{\textit{Adult Population}} \times 100\%$$

$$\therefore LFPR = \frac{34}{51} \times 100\% = 66.67\%$$

&, total employment rate (ER):

$$ER = \frac{\textit{Employed}}{\textit{Labor Force}} \times 100\%$$

$$ER = \frac{16}{34} \times 100\% = 47.05\%$$

(b)

➔ **Marginally Attached Workers**:

These are individuals who are not currently in the labor force, want and are available for work, and have looked for a job in the past 12 months but not in the last 4 weeks. They are not counted as part of the official labor force.

➔ **Part-Time Workers Who Want Full-Time Jobs**:

These are individuals who are working part-time (less than 35 hours per week) because they are unable to find full-time employment or due to economic reasons such as reduced hours by employers.

(c)

➔ Types of Unemployment

1. **Frictional Unemployment**:
    - This occurs when individuals are temporarily between jobs or are entering the labor force for the first time.
    - Example: A recent graduate looking for their first job or a person transitioning to a new role.
2. **Structural Unemployment**:
    - This occurs when there is a mismatch between the skills of workers and the requirements of jobs available.
    - Example: Workers in a factory losing jobs due to automation.
3. **Cyclical Unemployment**:
    - This is caused by economic downturns or recessions when the demand for goods and services declines, leading to layoffs.
    - Example: Job losses in the construction industry during a housing market crash.
4. **Seasonal Unemployment**:
    - This occurs when jobs are only available during certain times of the year.
    - Example: Agricultural workers during harvest seasons or retail employees hired for the holiday shopping season.

Each type of unemployment impacts the economy differently and requires tailored policies for resolution.

SUPPLY & DEMAND

TOPIC:4

THE MARKET FORCES OF SUPPLY AND DEMAND

- ⇨ Supply and Demand are the forces that make market economies work!
- ⇨ Modern microeconomics is about supply, demand, and market equilibrium.

MARKETS AND COMPETITION

- ⇨ The terms supply and demand refer to the behaviors of people as they interact with one another in markets.
- ⇨ A market is a group of buyers and sellers of a particular good or service.
  - Buyers determine demand.
  - Sellers determine supply.

DEMAND

- ⇨ Demand is the quantity of a commodity that a consumer is willing and **able to buy**, at each possible price during a given period of time.

There are Two types:

1. Individual Demand.
2. Market Demand.

Individual Demand

- ⇨ Refer to the quantity of a commodity that **a consumer** is willing and able to buy, at each possible price during a given period of time.

Market DEMAND

➾ Refer to the quantity of a commodity that **all consumers are** willing and a buy, at each possible price during a given period of time.

Determinants of Demand

1. **Price of the given commodity:** There exists an inverse relationship between price and quantity demanded. It means, as price increase, quantity demanded falls due to decrease in the satisfaction level of consumers,
2. **Price of related goods:** Demand for the given commodity is also affected by change in prices of the related goods.

Related goods are two types:

a. **Substitute goods:** Substitute goods are those goods which can be used in place of one another for satisfaction of a particular want, **like tea and coffee**.
An increase in the price of substitute leads to an increase in the demand for given commodity and vice-versa.

b. **Complementary goods**: Complementary goods are those goods which are used together to satisfy a particular want, like tea and sugar.
An increase in the price of complementary goods to a decrease in the demand for given commodity and vice-versa

3. **Income of the consumer:** The effect of change in income on demand depends on the nature of the commodity under consideration.

- If the given commodity is a **normal good**, then an increase in income leads to rise in this demand, while a decrease in income reduces the demand.

- If the given commodity is an **inferior good**, then an increase reduces the demand, while a decrease in income leads to rise in demand.

- **For examples**: Suppose, income of a consumer increase. As a result, the consumer reduces consumption of toned milk and increase consumption of full cream milk. In this case, " toned milk" is an inferior good and " full cream milk" is a normal good.

4. **Tastes and preferences**: Tastes and preference of the consumer directly influence the demand for a commodity
5. **Expectation of change in the price in future:** If the price of a certain commodity is expected to increase in near future, then people will buy more of that commodity than what they normally buy. There exists a direct relationship between expectation of change in the prices in future and change in demand in the cent period.

Law of Demand

⇨ **Law of Demand:** Law of demand states the inverse relationship between price and quantity demanded, keeping other factors constant (ceteris paribus).

- Price ⬆ Quality demanded ⬇
- Price ⬇ Quality demanded ⬆
- **Ceteris paribus**: Other things being constant or unchanged.

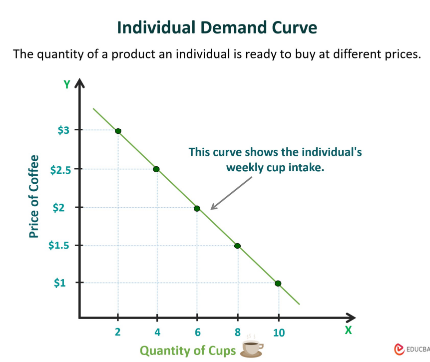
Demand Schedule

⇨ The demand schedule is a table that shows the relationship between the price of the good and the quantity demanded.

Demand Curve

⇨ The demand curve is a graphical representation of the demand schedule.

1. Individual Demand Curve:



2. Market Demand Curve:

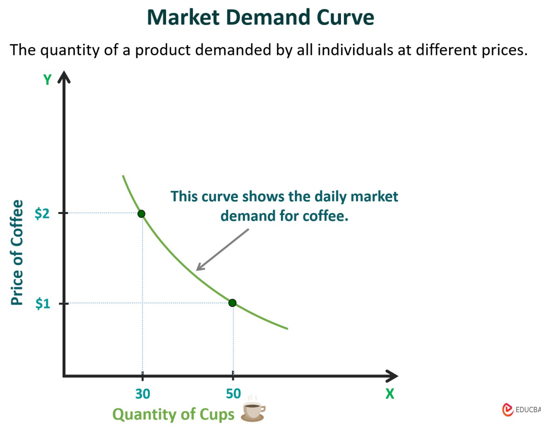


Fig1

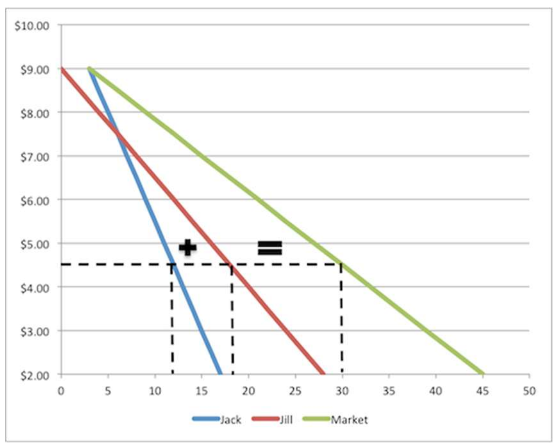


Fig 2

Shift in Demand Curve

1. **Increase in Demand:** Refers to a rise in demand of a commodity caused due to any factor other than the own price of the commodity.

Price (TK)	Demand (Units)
30	100
30	150

Fig: Increases in Demand, Rightward Shift

2. Decrease in Demand: Refers to a fall in the demand of a commodity caused due to any factor other a than the own price of the commodity

Price (TK)	Demand (Units)
30	100
30	80

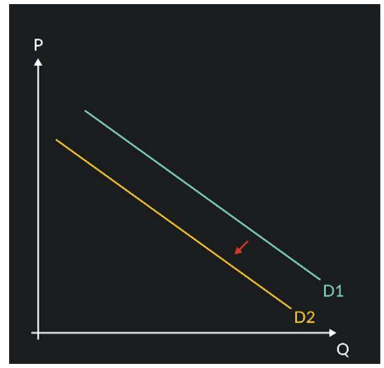


Fig: Decrease in Demand, Leftward Shift

Movement in Demand Curve

1. **Expansion in Demand:** Expansion in Demand Refers to a **rise in** the quantity demanded due to a **fall in the price of commodity**, other factors remaining Constant.

- It leads to a **downward movement** along the same demand curve.
- Also known as "Extension in Demand" or "**Increase in Quantity Demanded**"

Price (TK)	Demand (Units)
30	100
15	150

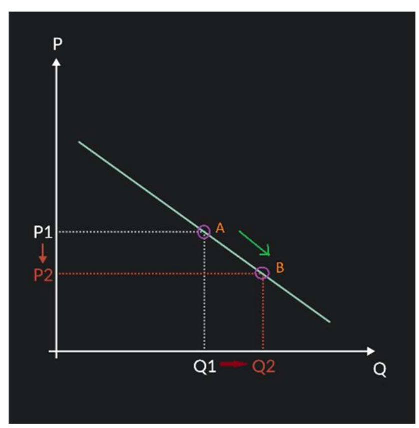


Fig: Increases in Demand, **downward movement**

2. **Contraction in Demand:-**Refers to a **fall** in the quantity demanded due to a **rise in the price** of commodity, other factors remaining constant.

- It leads to an **Upward movement** along the same demand curve.
- Also known as " Contraction in Demand" or "**Decrease in Quantity Demanded**"

Price (TK)	Demand (Units)
30	100
50	50

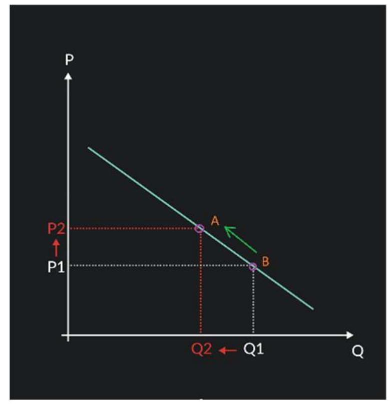


Fig: Decrease in Demand,
Upward movement

Reasons for Shift in Demand Curve

- Change in price of substitute goods.
- Change in price of complementary goods.
- Change in income of consumers.
- Change in tastes and preferences.
- Expectation of change in price in future.
- Change in population.
- Change in distribution of income.
- Change in season.

Reasons for Movement in Demand Curve

- **Extension of demand** (increase in quantity demanded) when the price falls.
- **Contraction of demand** (decrease in quantity demanded) when the price rises.

SYPPLY

⇨ Supply refers to **quantity of a commodity** that a firm is willing and able to offer for sale at a given **price** during a given **period of time**

There are Two types:

3. Individual Supply.
4. Market Supply.

Individual Supply

⇨ Refer to the quantity of a commodity that **an individual firm** is willing and able to offer for sale at a given **price** during a given **period of time**

Market Supply

⇨ Refer to the quantity of a commodity that **all firm** are willing and able to offer for sale at a given **price** during a given **period of time**

Determinants of Supply

1. **Price of the given commodity:** Price of a commodity and its supply are **directly** related. It means, as **price increases**, the quantity supplied of the given commodity **also rises** and vice-versa.

   Price⬆ Supply⬆

2. **Price of other goods:**

   Let, a supplier supply two goods Q1, Q2 at price P1, P2. Now if the price of Q1 increase the supplier increase the supply of Q1 goods and decrease the supply of Q2 goods and vice-versa.

3. **Price of the factors of production or inputs:** Price of the factors of production or inputs (like, labor, capital, raw material, etc.) used in the process of production constitute the cost of production of the commodity.

    a) If the prices of all or any of these factors or inputs **increases**, **the cost of production also increases**. This **decreases the profitability**. As a result, seller reduces the supply of the commodity.
    b) **Decreases** in prices of factors of production or inputs, increases the supply due to fall in cost of production and subsequent **rise in profit margin**.

4. **States of Technology:**
    a) Technological changes influence the supply of a commodity. **Advanced** and **improved technology** **reduces the cost** of production, which **raises the profit margin**. It induces a seller to increase the supply.
    b) Technological **degradation** or **complex** and **outdated technology** will **increase the cost of production** and it will lead to decrease in supply.

5. **Expectation of change in the price in future:** If sellers expect a rise in price in near future, then current market supply will decrease in order to raise the supply in future at higher prices. However, if the sellers fear that the prices will fall in the future, then they will increase the present supply to avoid losses in future.

6. **Government Policy (Taxes & Subsidies):** Increase in taxes raises the cost of production and, thus, a reduces the supply, due to lower profit margin. On the other hand, tax concessions increases supply as they make it more profitable for the firms to supply goods.

7. **Means of transportation and communication:** Proper infrastructural development, like improvement in the means of transportation and communication, help in maintaining adequate supply of the commodity

Law of Supply

⇨ The law of supply states that, other things equal, the quantity supplied of a good, rises when the price of the good rises.

- Price ⬆ Supply ⬆
- Price ⬇ Supply ⬇
- **Ceteris paribus**: Other things being constant or unchanged.

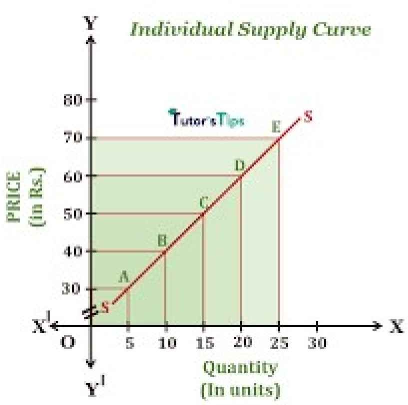
Supply Schedule

⇨ The Supply schedule is a table that shows the relationship between the price of the good and the quantity **supplied**.

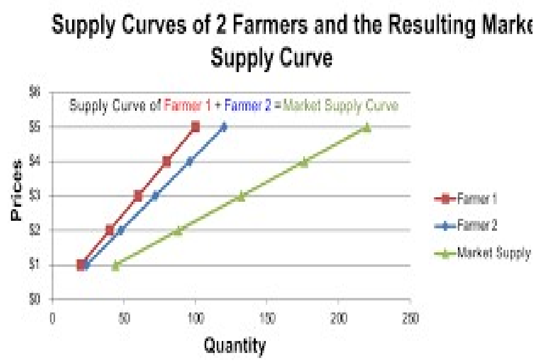
Supply Curve

⇨ The Supply curve is a graphical representation of the Supply schedule.

1. Individual Supply Curve:



2. Market Supply Curve:



2. Market Supply Curve:

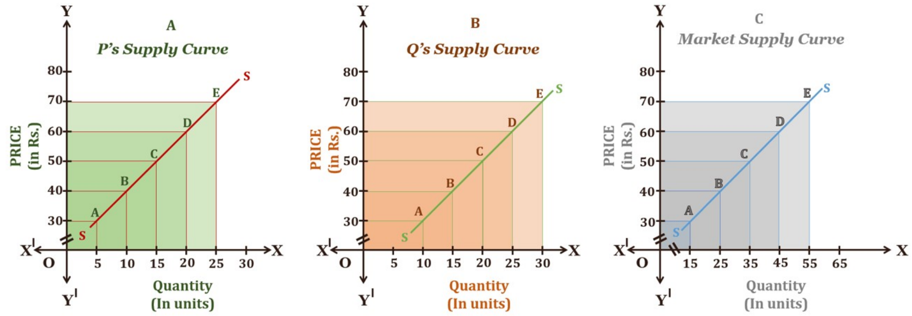


Fig 2

Shift in Supply Curve

1. **Increase in Supply:** Refers to a rise in supply of a commodity caused due to any factor other than the own price of the commodity.

Price (TK)	Quantity (Units)
5	100
5	200

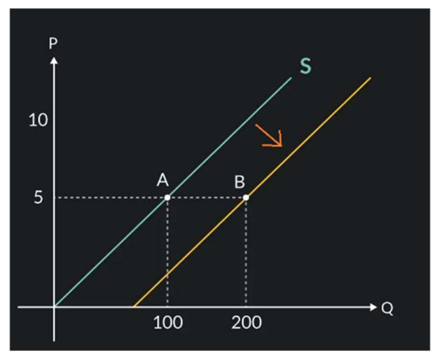


Fig: Increases in Supply, Rightward Shift

2. **Decrease in Supply:** Refers to a fall in the Supply of a commodity caused due to any factor other a than the own price of the commodity

Price (TK)	Quantity (Units)
5	100
5	25

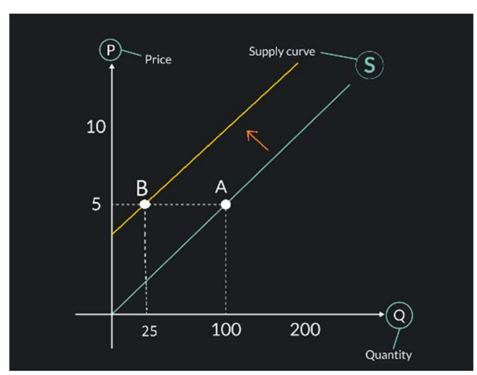


Fig: Decrease in Supply, Leftward Shift

Movement in Supply Curve

1. **Expansion in Supply:** Expansion in Supply Refers to a **rise in** the quantity demanded due to a **increase in the price of commodity**, other factors remaining Constant.

   - It leads to a **upward** **movement** along the same supply curve.
   - Also known as "Extension in Supply" or "**Increase in Quantity Supplied**"

Price (TK)	Quantity (Units)
5	100
10	200

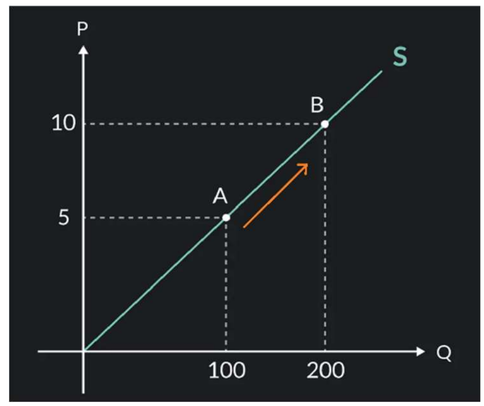


Fig: Increases in Supply,
Upward movement

2. **Contraction in Supply:-**Refers to a **fall** in the quantity Supplied due to a **fall in the price** of commodity, other factors remaining constant.

   - It leads to an **Downward** **movement** along the same Supply curve.
   - Also known as " Contraction in Supply" or "**Decrease in Quantity Supplied.**

Price (TK)	Demand (Units)
5	100
2.5	50

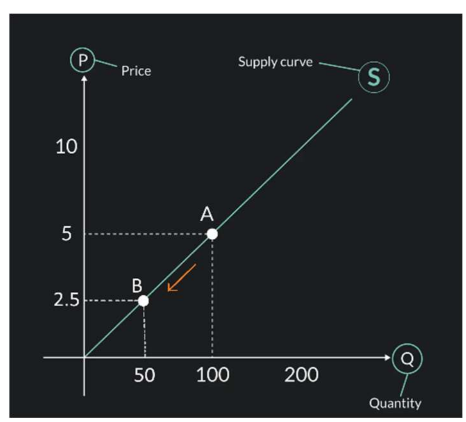


Fig: Decrease in Supply,
Downward movement

Reasons for Shift in Supply Curve

- **Changes in production costs** (e.g., labor, raw materials).
- **Technological advancements** improving efficiency.
- **Taxes and subsidies** affecting production costs.
- **Number of suppliers** entering or exiting the market.
- **Natural events or disasters** disrupting supply.
- **Government regulations** increasing or decreasing production constraints.
- **Price expectations** of future goods.
- **Availability of resources** or inputs.

Reasons for Movement in Supply Curve

- **Extension of supply** (increase in quantity supplied) when the price rises.
- **Contraction of supply** (decrease in quantity supplied) when the price falls.

SUPPLY AND DEMAND TOGETHER (*Equilibrium*)

⇨ ***Equilibrium*** refers to a situation in which the price has reached the level where **quantity supplied equals quantity demanded.**

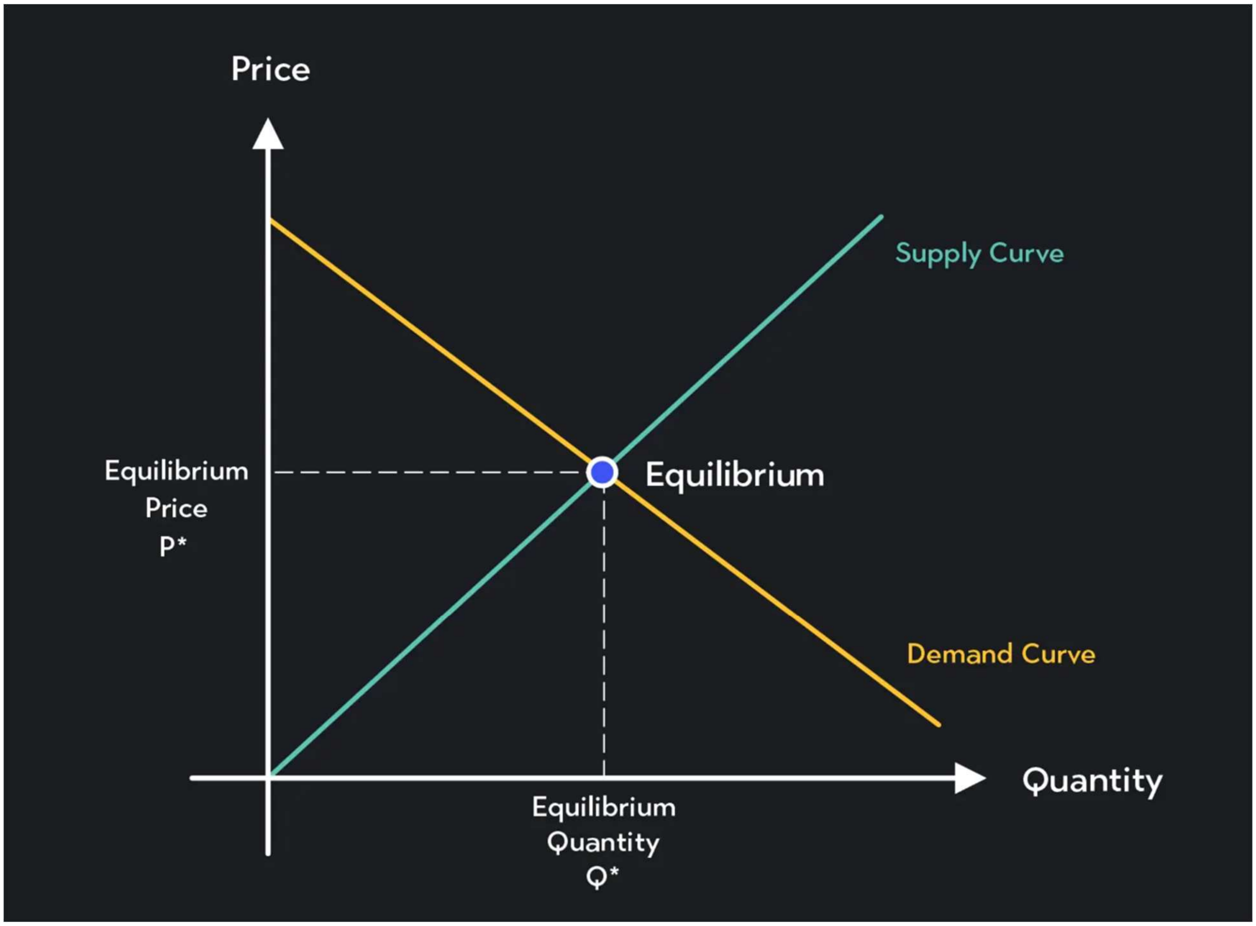
Equilibrium Price

- The price that balances quantity supplied and quantity demanded.
- On a graph, it is the price at which the **supply and demand curves intersect.**

Equilibrium Quantity

- The quantity supplied and the quantity demanded at the equilibrium price.
- On a graph it is the quantity at which the **supply and demand curves intersect.**

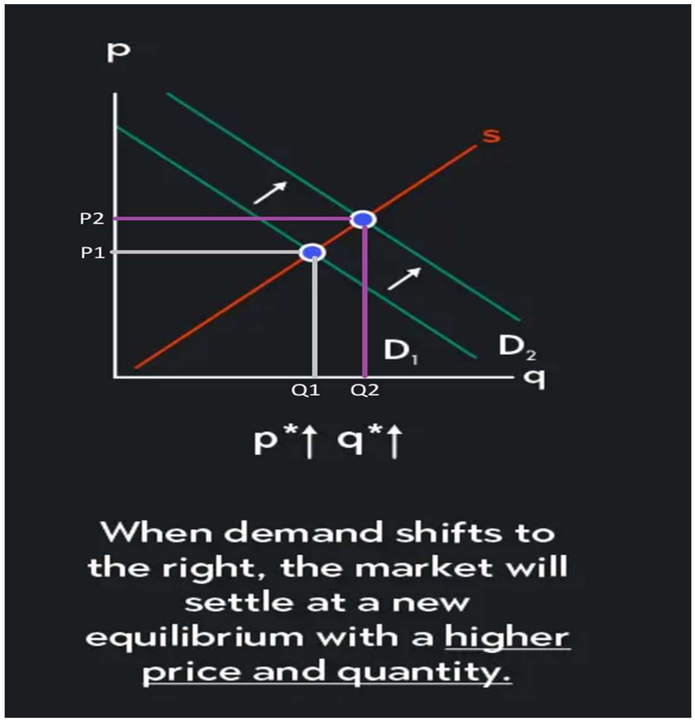
The Equilibrium of Supply and Demand



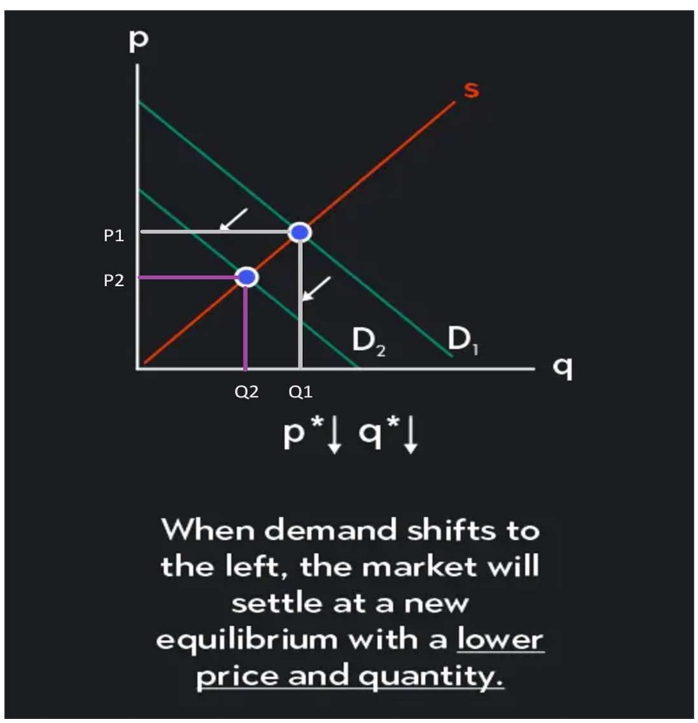
Steps To Analyzing Changes in Equilibrium

1. Decide whether the event shifts the supply or demand curve (or both).
2. Decide whether the curve(s) shift(s) to the left or to the right.
3. Use the supply-and-demand diagram to see how the shift affects equilibrium price and quantity.

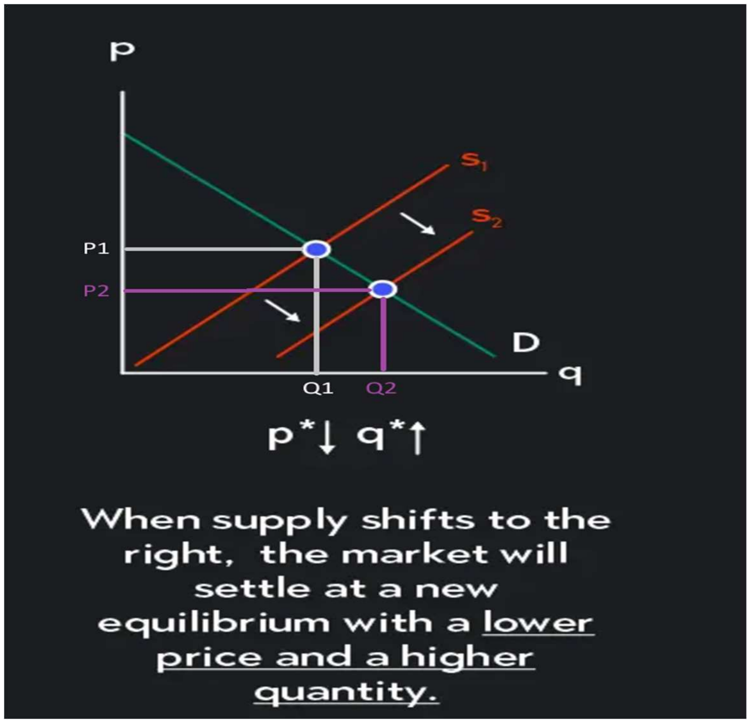
How an Increase Demand Affects the Equilibrium



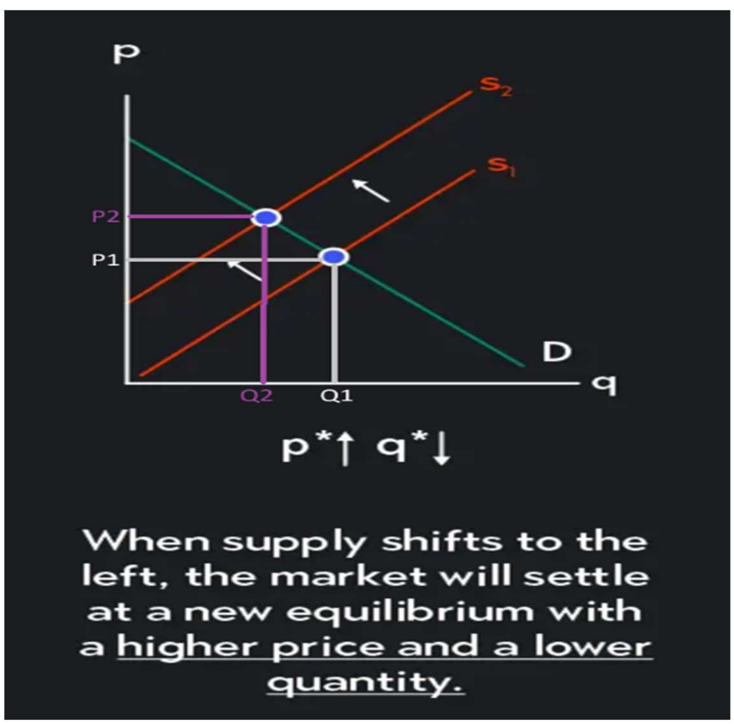
How a Decrease Demand Affects the Equilibrium



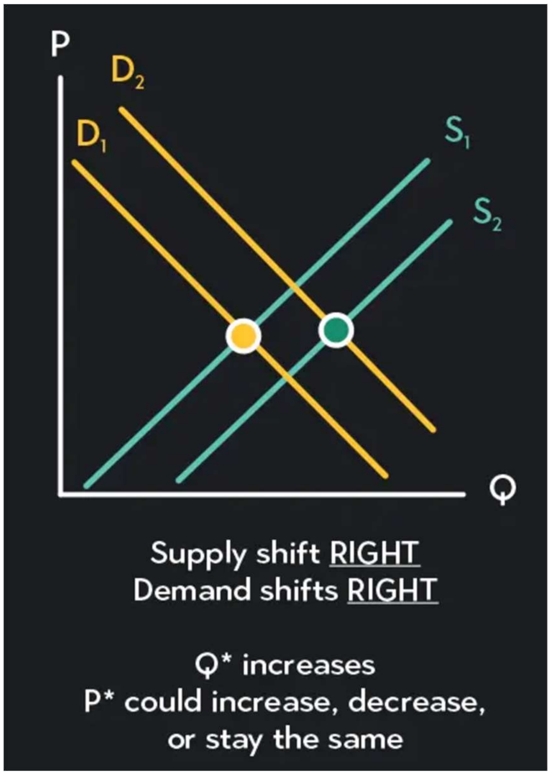
How an Increase Supply Affects the Equilibrium

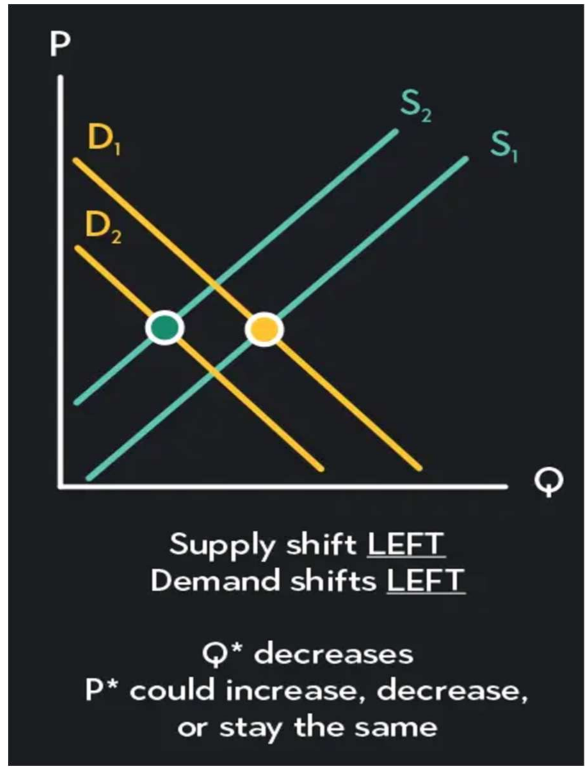


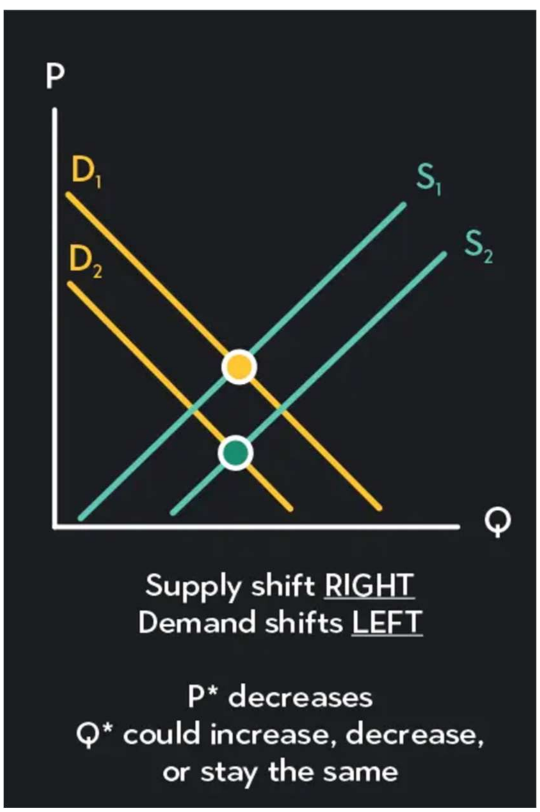
How a Decrease Supply Affects the Equilibrium

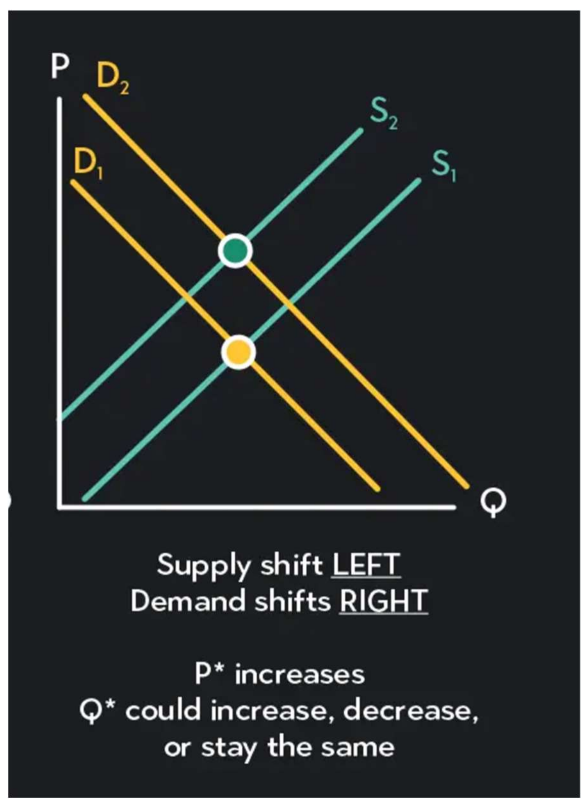


A Shift in Both Supply and Demand









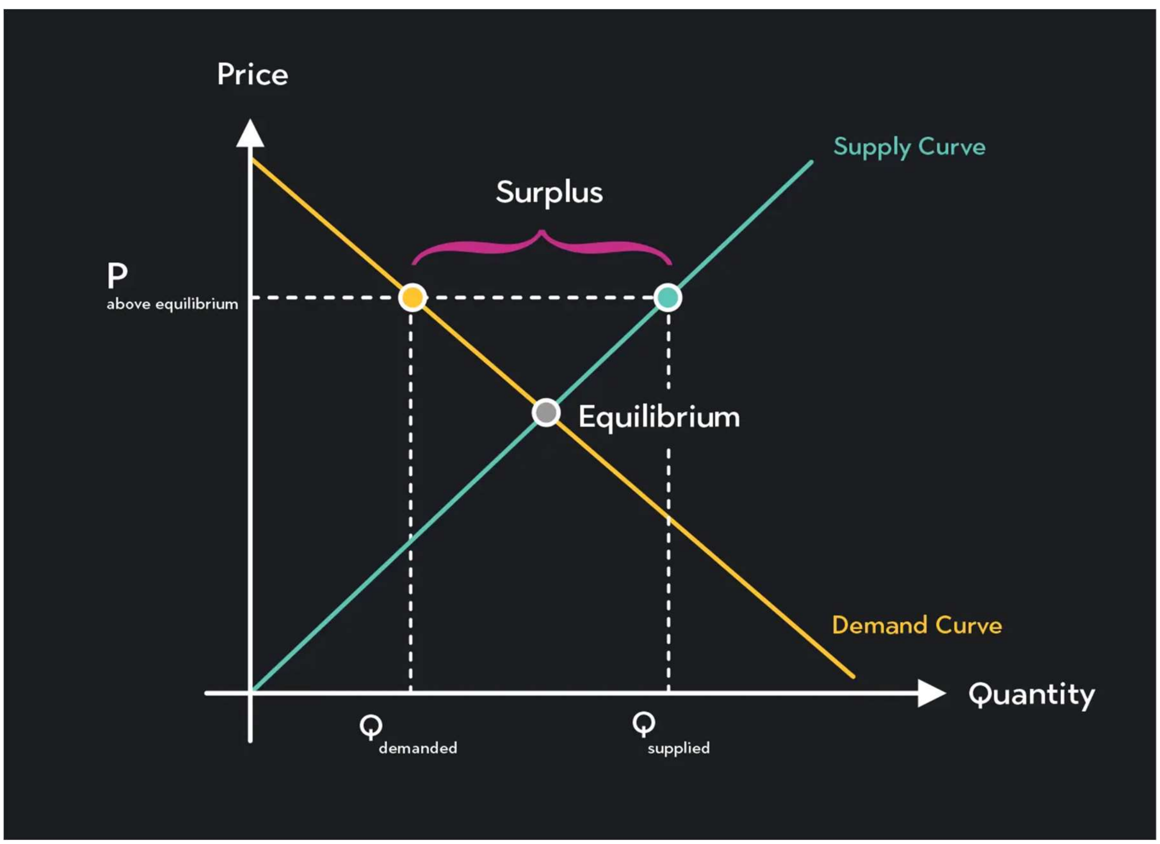
What Happens to Price and Quantity when Supply or Demand Shifts

	No Change in Supply	An Increase in Supply	A Decrease in Supply
No Change in Demand	*P* same *Q* same	*P* down *Q* up	*P* up *Q* down
An Increase in Demand	*P* up *Q* up	*P* ambiguous *Q* up	*P* up *Q* ambiguous
A Decrease in Demand	*P* down *Q* down	*P* down *Q* ambiguous	*P* ambiguous *Q* down

Surpluses and Shortages

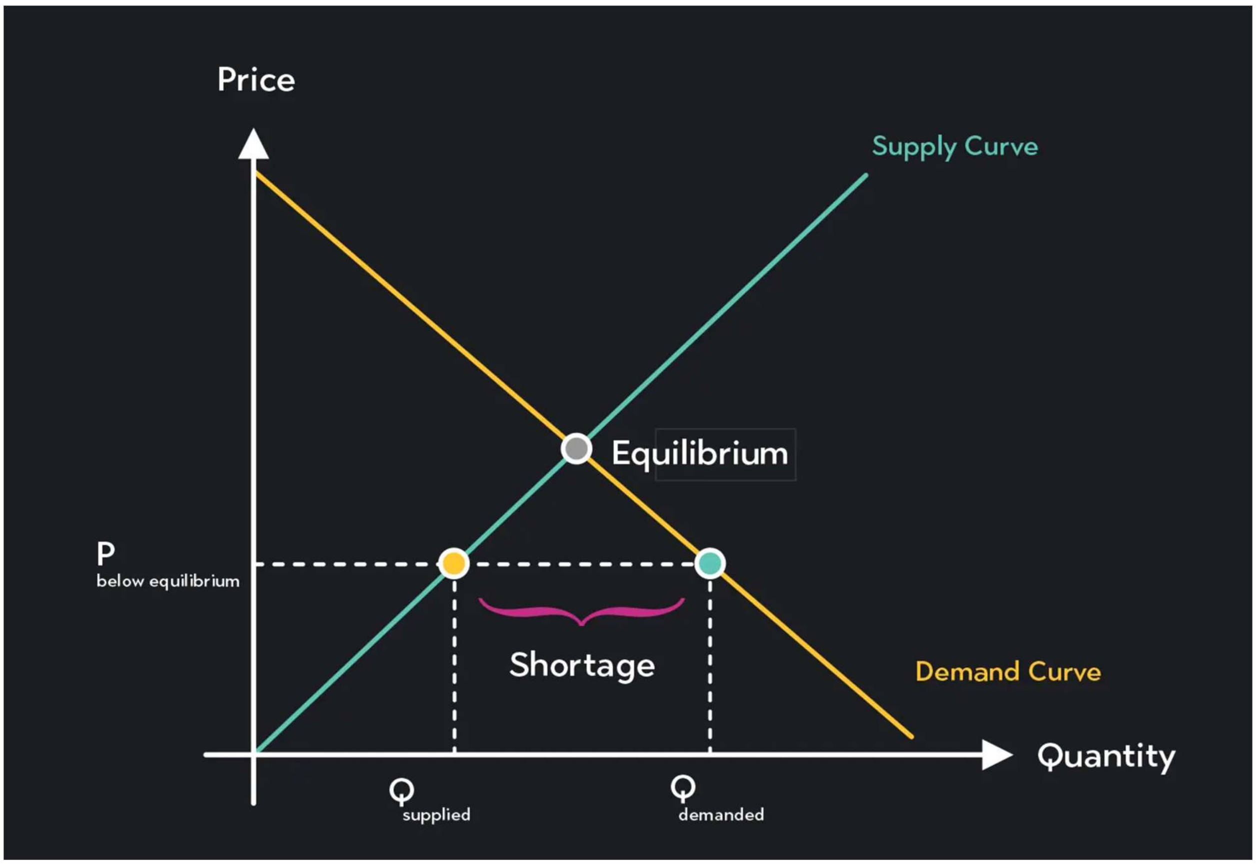
- ***Surplus***

⇨ **When price > equilibrium price, then quantity supplied > quantity demanded.**
⇨ There is excess supply or a surplus.
⇨ Suppliers will lower the price to increase sales, thereby moving toward equilibrium.



- ***Shortage***

⇨ **When price < equilibrium price, then quantity demanded > the quantity supplied.**
⇨ There is excess demand or a shortage.
⇨ Suppliers will raise the price due to too many buyers chasing too few goods, thereby moving toward equilibrium.



#Problem 1:

1. Following Table shows the demand schedule of Good X.

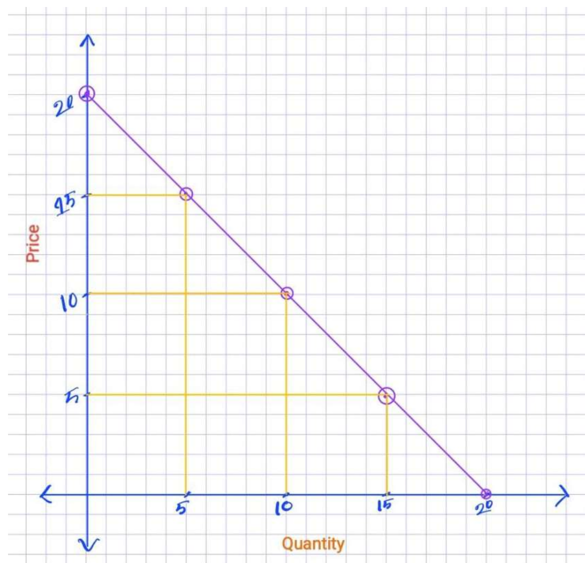
Price of X	Quantity demanded of X
20	0
15	5
10	10
5	15
0	20

a. Draw the demand curve for Good X.

b. Suppose Good Y is a substitute for good X. On such case what will be the change in the demand curve of good X, Show that.

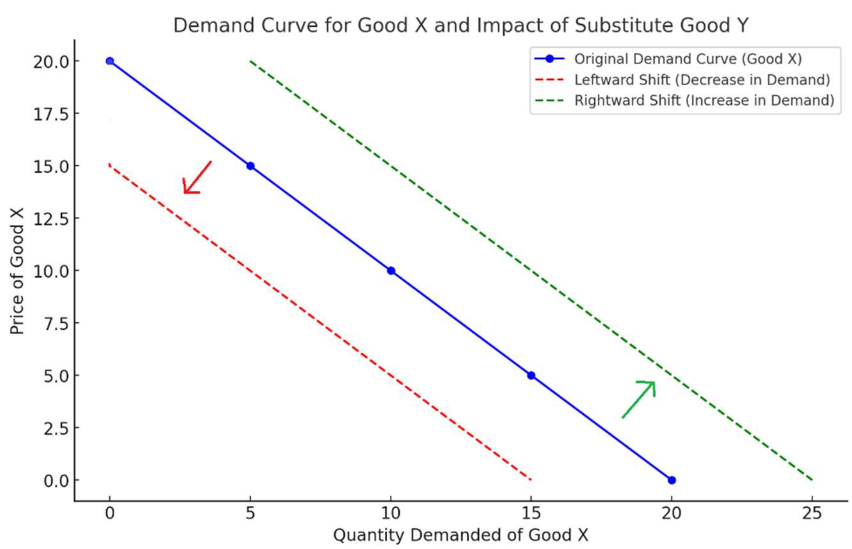
(a)

Now the Demand curve is:



(b)

- If the price of Good Y decreases, consumers may shift to Good Y, reducing the demand for Good X. This causes the **demand curve for Good X to shift leftward**.
- Conversely, if the price of Good Y increases, consumers may switch to Good X, increasing its demand. This causes the **demand curve for Good X to shift rightward**.



#Probem 2:

2. Following Table shows the supply and demand schedule of sweater.

Price	Quantity Demanded	Quantity Supply
20	0	25
15	5	20
8	10	10
5	15	7
0	20	0

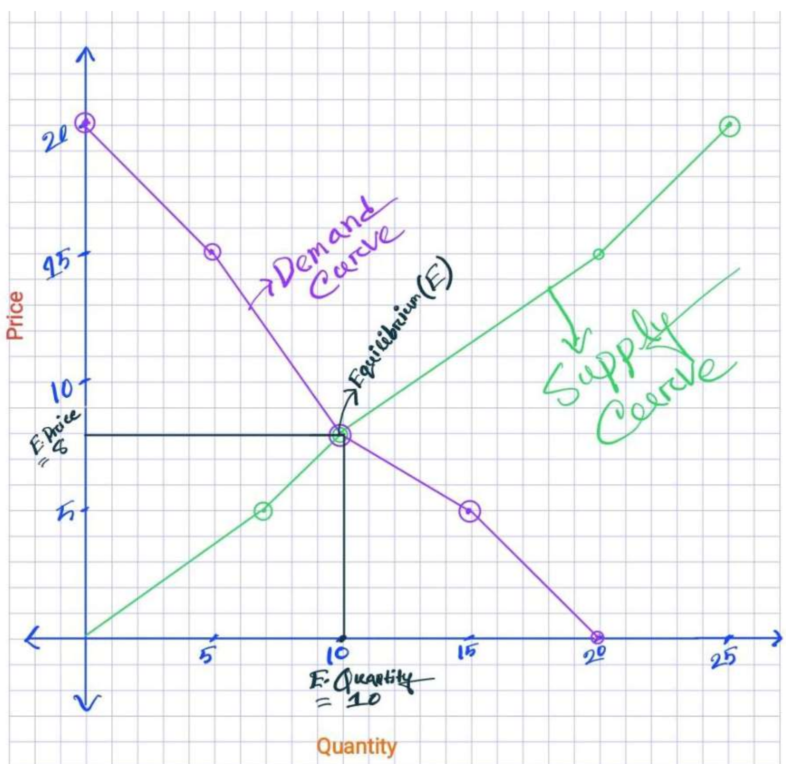
a. Draw the demand and supply curve for Good X.

b. Identify the equilibrium point and what is the equilibrium price and quantity in the sweater market?

c. Suppose in winter season the weather is **extremely** cold. In addition to that due to extreme cold **some** of the sheep got died which is the important resources for sweater production. How do these two events affect the market equilibrium (demand and supply). How will it change the equilibrium price and quantity?

(a)

Now the Demand & Supply curve is:



(b)

We Know,

The **equilibrium** is the point where $Q_D = Q_S$

From the table, at P=8 $Q_D = Q_S = 10$.

- **Equilibrium Price**: P=8
- **Equilibrium Quantity**: Q=10

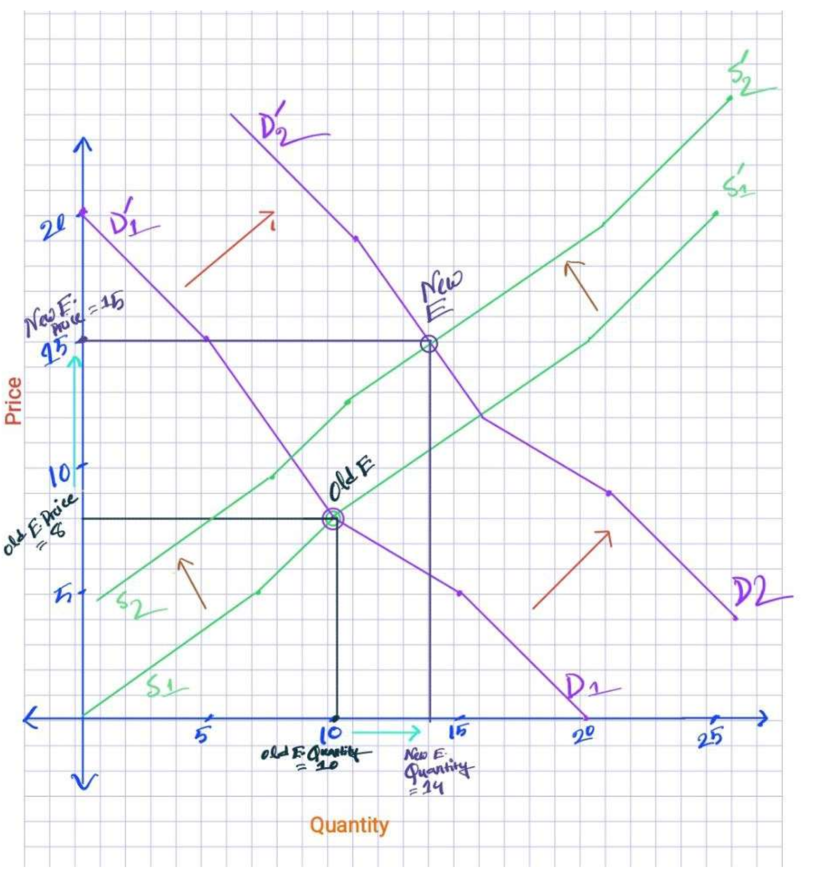
(c)

➔ **For Extreme cold weather (increase in demand)**:

- Increased demand for sweaters as more people needs them for warmth.
- The **demand curve shifts rightward**.

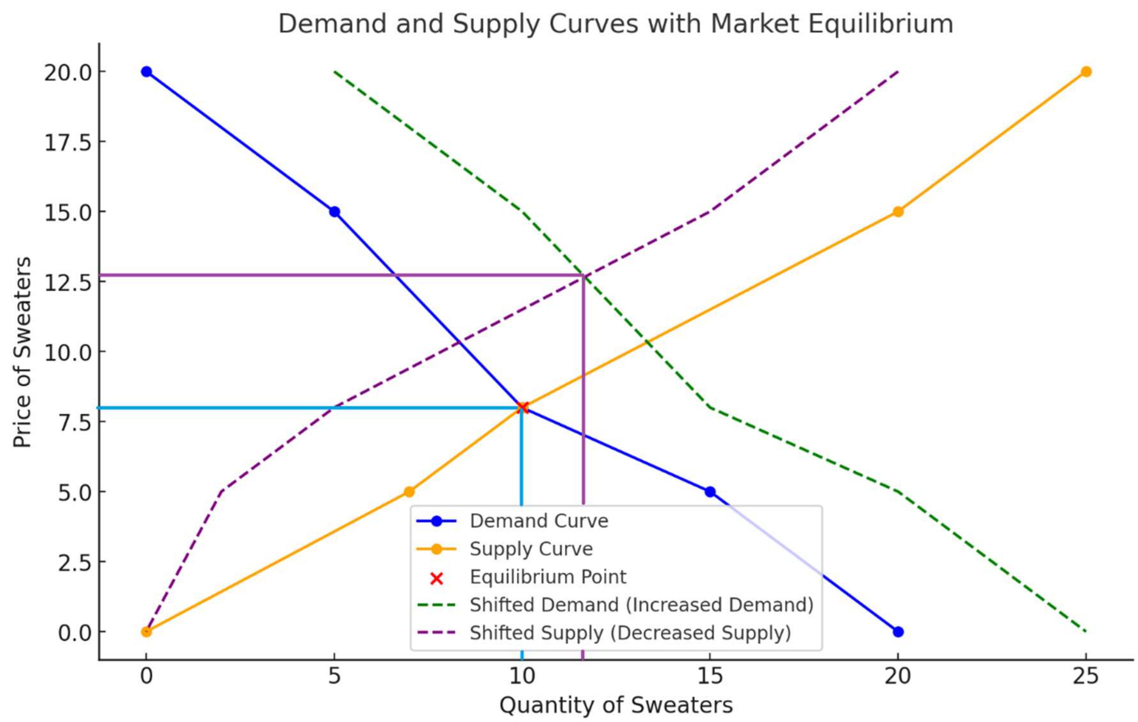
➔ **For Reduction in sheep population (decrease in supply)**:

- Fewer resources for sweater production.
- The **supply curve shifts leftward**.



Effects on the Equilibrium:

- The **equilibrium price** increases due to the combined effects of higher demand and lower supply.
- The **equilibrium quantity** also increases but not so much due to reduced production capacity, despite increased demand.



Problem 3:

a) Consider **a market of cake**. Suppose suddenly the **price of pastry increase** on the other hand due the **price of butter (one of the ingredients) decrease**. In such case, **how these events will affect the market (demand and supply) equilibrium of cake?** Explain with diagram.

b) Explain any four determinants that can shift the demand curve with relevant diagram

➔Solution:

(a)

1. Impact on Demand:

- When the price of pastries increases, some consumers may shift their preference to cakes as a substitute good. This will increase the demand for cakes.
- An increase in demand will shift the **demand curve for cakes to the right**.

2. Impact on Supply:

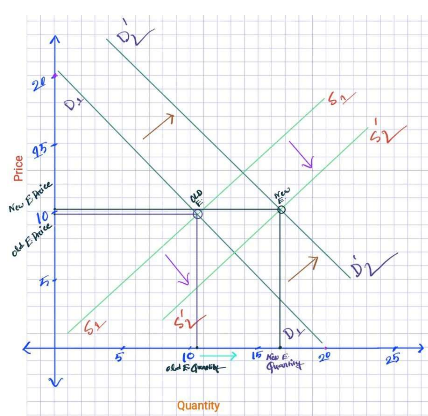
- A decrease in the price of butter, which is a key ingredient in cakes, reduces the production cost. This encourages suppliers to produce more cakes.
- An increase in supply will shift the **supply curve for cakes to the right**.

3. Equilibrium Effect:

- The combined effect of increased demand and supply may lead to a **higher equilibrium quantity of cakes**. However, the impact on the price depends on the magnitude of the shifts in demand and supply:
  - If the demand increase is larger than the supply increase, the equilibrium price will rise.
  - If the supply increase is larger than the demand increase, the equilibrium price will fall.

Diagram:

- A standard supply and demand diagram can illustrate the shifts:
  - The **demand curve** shifts right due to increased demand for cakes.
  - The **supply curve** shifts right due to reduced production costs from cheaper butter.
  - The new equilibrium shows the effect on price and quantity.

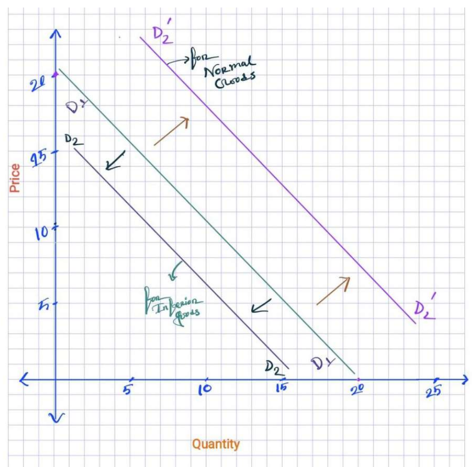


(b)

➔ Determinants that Can Shift the Demand Curve

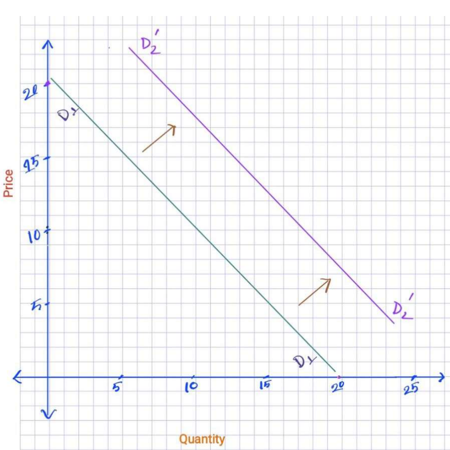
1. **Income of Consumers:**
   - When consumer income increases, the demand for normal goods (like cakes) tends to rise, shifting the demand curve to the right.
   - For inferior goods, higher income may reduce demand, shifting the curve to the left.

Diagram: A demand curve shifting right with an income increase.



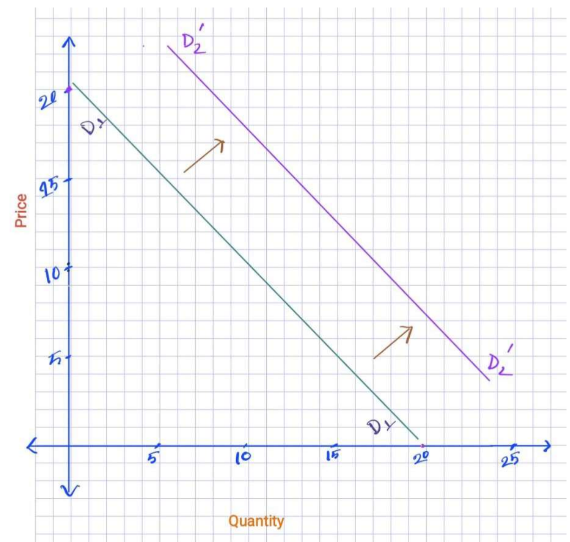
2. **Prices of Related Goods:**
   - **Substitutes:** If the price of a substitute good (e.g., pastries) rises, the demand for cakes increases, shifting the curve to the right.
   - **Complements:** If the price of a complement (e.g., coffee) decreases, the demand for cakes also increases.

Diagram: Shift due to a change in the price of a related good.



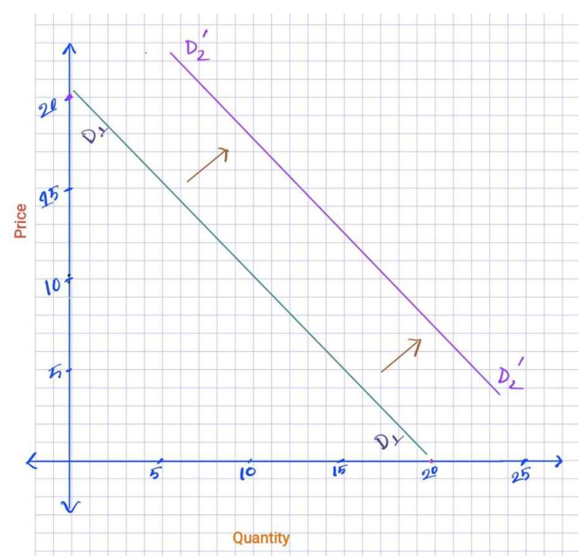
3. **Consumer Preferences:**
    - A change in preferences in favor of cakes (e.g., due to a trend or health campaign) will increase demand, shifting the curve to the right.

Diagram: Demand curve shifts right due to increased consumer preference.



4. **Population Size or Market Size:**
    - An increase in the number of consumers (e.g., population growth) leads to higher demand, shifting the demand curve to the right.

Diagram: Demand curve shifts right due to a larger market size.



MONEY AND MONETARY POLICY

TOPIC:5

Money and Monetary Policy

- ⇨ Money refers to assets commonly used in an economy to buy goods and services.

The Functions of Money

- ⇨ **Money has three functions in the economy:**
  - **Medium of exchange**
  - **Unit of account**
  - **Store of value**

Medium of exchange

- An item buyers use to pay sellers for goods and services.
- Anything widely accepted as payment.

Unit of account & Store of value

- **Unit of Account**: A way to measure and compare prices.
- **Store of Value**: Something people can save to use in the future.

Concept of money

- **Liquidity**: How easily an asset can be turned into cash or money.

The Kinds of Money:

- **Commodity Money**: Money with intrinsic value, like gold or silver.
- **Fiat Money**: Money with no intrinsic value, used because of government order (e.g., coins, currency, bank deposits).
- **Currency**: Paper bills and coins held by the public.
- **Demand Deposits**: Bank account balances that can be accessed anytime by check or electronic transfer.

BANKS AND THE MONEY SUPPLY

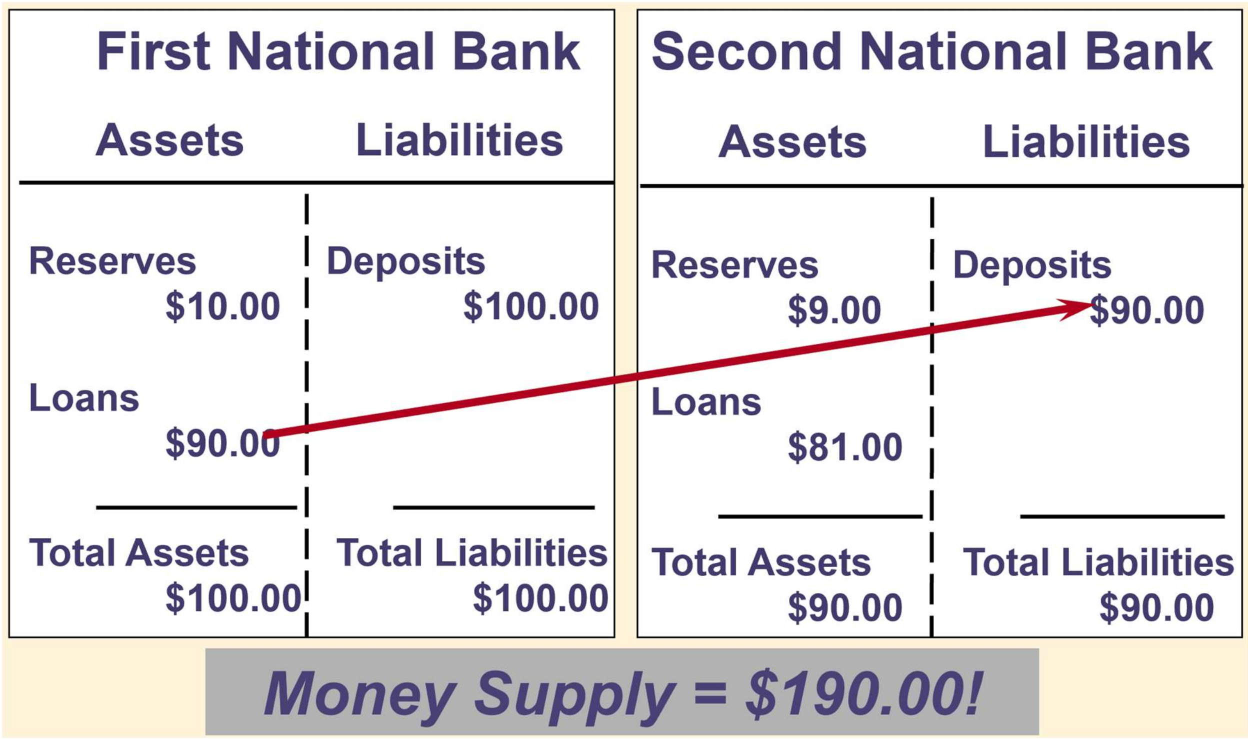
- **Reserves:** Money deposited in banks that is not loaned out.
- **Fractional-Reserve Banking**: Banks keep part of deposits as reserves and lend out the rest.
- **Reserve Ratio**: The percentage of deposits banks keep as reserves.

Money Creation with Fractional-Reserve Banking

- **Loans and Money Supply**: Loans made from reserves increase the money supply.
- **Money Supply Factors**: Influenced by deposits and bank loans.
- **Deposits**: Recorded as both assets and liabilities by banks.
- **Reserve Ratio**: The fraction of deposits banks must hold as reserves.
- **Loans**: Considered an asset for the bank.

This T-Account shows a bank that

- **Bank Functions**:
  - Accepts deposits.
  - Keeps a portion as reserves (e.g., 10%).
  - Lends out the rest.



- **Bank Loans and Deposits**: Loans from one bank are usually deposited into another.
- **Cycle of Lending**: Creates more deposits and reserves for lending.
- **Money Supply Growth**: Loans from reserves increase the money supply.

The Money Multiplier

- **Money created in economy = Money multiplier** × **Initial reserves**
- **Money multiplier** = **1 / Reserve ratio (R)**

⇨ **If Reserve ratio (R)** = **20% or 1/5**

Then, Money multiplier = 5

∴ **Money created = 5 × Initial reserves**

The Central Bank's Tools of Monetary Control

- **Central Bank tools:**
  - Open-market operations
  - Changing reserve requirement
  - Changing discount rate
- **Open-Market Operations:**
  - Buy bonds → money supply increases
  - Sell bonds → money supply decreases
- **Reserve Requirements:**
  - Minimum reserves banks must hold
  - Increase reserve requirement → money supply decreases
  - Decrease reserve requirement → money supply increases
- **Discount Rate:**
  - Interest rate for bank loans from the Central Bank
  - Increase discount rate → money supply decreases
  - Decrease discount rate → money supply increases

Problem 1:

a) Explain the circular flow of money with diagram.

b) What is money multiplier?

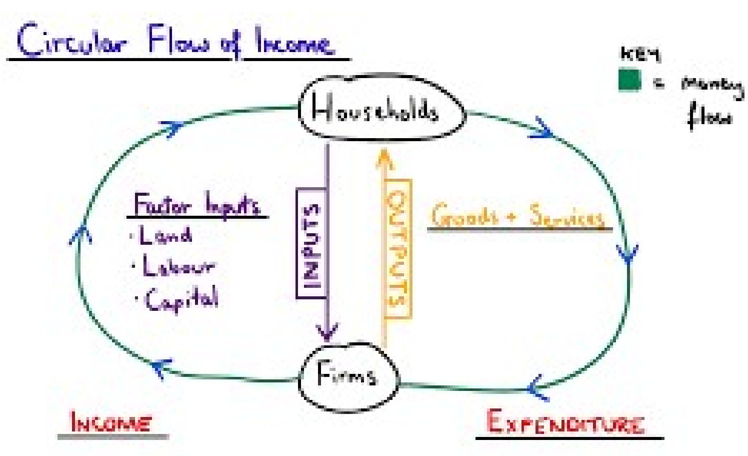
c) Explain central bank tools to control money supply.

(a)

➔**Circular Flow of Money:** The circular flow of money represents the movement of money, goods, and services in an economy. It involves households, businesses, government, and the financial sector. Here's a simplified explanation:

- **Households** provide labor and consume goods and services.
- **Businesses** produce goods and services and pay wages to households.
- **Financial sector** (banks) allows savings and investments to circulate.
- **Government** collects taxes and provides public goods and services.

Diagram:



(b)

➔Money Multiplier:

The money multiplier is the amount of money the banking system creates with each unit of reserves. It is the reciprocal of the reserve ratio:

Formula:

⇨ Money created in economy = Money multiplier × Initial reserves

⇨ Money multiplier = 1 / Reserve ratio (R)

- **Example:**
  If the reserve ratio is 20% (or 0.2), the money multiplier is 5. This means that for every dollar of reserves, the banking system can create 5 dollars of money.

(c)

➔ Central Bank Tools to Control Money Supply:

The Central Bank uses several tools to influence the money supply in the economy:

1. **Open-Market Operations:**
   - Buying bonds → increases the money supply.
   - Selling bonds → decreases the money supply.
2. **Changing Reserve Requirements:**
   - Increasing the reserve requirement → reduces the money supply.
   - Decreasing the reserve requirement → increases the money supply.
3. **Changing the Discount Rate:**
   - Increasing the discount rate → reduces the money supply.
   - Decreasing the discount rate → increases the money supply.

These tools are used to manage inflation, control interest rates, and stabilize the economy.